MIL-C-915E
1 August 1972
SUPERSEDING
MIL-C-00915D(SHIPS)
20 March 1972
MIL-C-915C
15 March 1968
(See 6.7 and 6.8)

MILITARY SPECIFICATION

CABLE AND CORD ELECTRICAL, FOR SHIPBOARD USE

GENERAL SPECIFICATIONS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical cable, and cord for shipboard applications.

1.2 Classification. Cables and cords covered by this specification shall be classified as watertight and non-watertight constructions and further classified for flexing and non-flexing service for power, lighting, control, communications, instrumentation and electronic applications.

1.2.1 Watertight, non-flexing service.

Power and Lighting:

Type	DRW	Two conductors	3000 volts
	DSGU	Two conductors	1000 volts
	FSGU	Four conductors	1000 volts
	MDU	Nineteen conductors	600 volts
	SRW	Single conductor	3000 volts
	SSGU	Single conductor	1000 volts
	TRW	Three conductors	3000 volts
	TSGU	Three conductors	1000 volts
	5KVTSGU	Three conductors	5000 volts
	6SGU	Six conductors	1000 volts
	7SGU	Seven conductors	1000 volts

Control:

Type	MSCU	7 through 91 conductors	1000 volts

Electronic, Communication and Instrumentation:

Type	ECM	56 single conductors plus 8 shielded pairs	600 volts
	MSPW	59 conductors (8 shielded pairs, 3 shielded triads, 34 shielded singles)	
	PBTMU	5, 15 and 30 pairs	
	TCJU	1 pair	
	TCJX	3, 7 and 12 pairs	
	TCKX	1, 3, 7 and 12 pairs	
	TCTU	1 pair	
	TCTX	3, 7 and 12 pairs	
	TSP	11 and 31 pairs	300 volts
	TTSU	1-1/2 through 60 pairs	300 volts
	1SMWU	70 shielded singles	
	1SWU	2, 14, 20 and 30 shielded singles	

FSC 6145

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2SWAU	3 through 61 shielded pairs	
2SWU	1 through 61 shielded pairs	
2WAU	40 pairs with overall shield	600 volts
3SWU	3 through 44 shielded triads	

1.2.2 Watertight, flexing service.

Power:

Type DSWS	Two conductors shielded	
MCSF	Four conductors (2-#6 and 2-#1 AWG)	600 volts

Control:

Type MWF	7 through 37 conductors	600 volts

Electronic, Communication and Instrumentation:

Type DLT	Four conductors	300 volts
DSS	Two conductors with overall shield	600 volts
FSS	Four conductors with overall shield	600 volts
S2S	Two conductor (concentric)	
TPU	Six pairs	
TSS	Three conductors with overall shield	600 volts
1SWF	Two shielded singles	
2SWF	3, 4 and 7 shielded pairs	
5SS	Five conductors (4 singles and one shielded single) with overall shield	600 volts
7SS	Seven conductors with overall shield	600 volts

1.2.3 Non-watertight, non-flexing service.

Power and Lighting:

Type DNW	Two conductors	1000 volts
DPS	Two conductors	600 volts
FNW	Four conductors	1000 volts
FPS	Four conductors	600 volts
TNW	Three conductors	1000 volts
TPS	Three conductors	600 volts
2SJ	Two conductors with overall shield	600 volts
3SJ	Three conductors with overall shield	600 volts
4SJ	Four conductors with overall shield	600 volts
7PS	Seven conductors	600 volts

Control:

Type MNW	7 through 44 conductors	1000 volts
MS	37 conductors with overall shield	300 volts
MU	14 conductors	300 volts

Electronic, Communication and Instrumentation:

Type MRI	Twisted pair and twisted triad	
MSP	59 conductors (8 shielded pairs, 3 shielded triads, 34 shielded singles)	
PI	3, 7, and 12 shielded pairs	
TPNW	1-1/2 through 40 pairs	300 volts
1SAU	44 shielded singles	
1SMU	5 shielded singles	
1SU	36 and 60 shielded singles	
1S50MU	16, 20, 40 and 70 shielded singles	
1S75MU	8 shielded singles	
2AU	40 pairs with overall shield	600 volts
2SJ	2 conductor with overall shield	600 volts
2SU	3 through 61 shielded pairs	
2U	10 through 60 pairs with overall shield	300 volts

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3SJ	Three conductors with overall shield	600 volts
3SU	3 through 44 shielded triads	
3U	3, 7 and 12 triads	
4SJ	Four conductors with overall shield	600 volts

1.2.4 Non-watertight flexing service.

Power and Lighting:

Type CVSF	Four conductors (3-#53 and 1-#33)	600 volts
DHOF	Two conductors sizes 3 through 400	600 volts
FHOF	Four conductors sizes 3 through 133	600 volts
JAS	Four conductors (2-#250 and 2-#6)	600 volts
SHOF	Single conductors sizes 3 through 800	600 volts
SSF	Single conductor	600 volts
THOF	Three conductors, sizes 3 through 600	600 volts
TRF	Single conductor	600 volts
TRXF	Single conductor	125 volts

Control:

# Type MHOF	7 through 61 conductors	600 volts

Electronic, Communication and Instrumentation

Type DCOP	Two conductors	300 volts
MCOS	Multi conductors-shielded	600 volts
MMOP	Five conductors	300 volts
TCOP	Three conductors	300 volts
TTOP	3 through 15 pairs	300 volts
TTRS	2 through 16 shielded pairs	300 volts
3SF	7 shielded triads	600 volts

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-R-175 - Resistance Wire.
TT-P-320 - Pigment, Aluminum: Powder and Paste, For Paint.
PPP-V-205 - Veneer, Paper Overlaid, Container Grade.

MILITARY

MIL-G-160 - Grips, Cable (Naval Shipboard Use).
MIL-C-572 - Cords, Yarns and Monofilaments, Organic Synthetic Fiber.
MIL-Y-1140 - Yarn, Cord, Sleeving, Cloth and Tape-Glass.
MIL-C-12000 - Cable, Cord and Wire, Electric, Packaging of.
MIL-L-17331 - Lubricating Oil, Steam Turbine (Non-Corrosive).
MIL-S-24235/1 - Stuffing Tube, Bulkhead, Pressureproof.
MIL-S-24235/2 - Packing Assemblies for Pressureproof Bulkhead Stuffing Tubes.

See Supplement-1 for applicable specification sheets.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM-B 3 - Soft or Annealed Copper Wire.
ASTM-B 8 - Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
ASTM-B 33 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.
ASTM-B 172 - Rope-Lay Stranded Copper Conductors Having Bunch-Stranded Members, for Electrical Conductors.
ASTM-B 173 - Rope-Lay Stranded Copper Conductors Having Concentric Members, for Electrical Conductors.
ASTM-B 174 - Bunch-Stranded Copper Conductors for Electrical Conductors.
ASTM-B 189 - Lead-Coated and Lead Alloy-Coated Soft Copper Wire for Electrical Purposes.
ASTM-B 193 - Resistivity of Electrical Conductor Materials, Test for.
ASTM-B 228 - Concentric-Lay-Stranded Copper Clad Steel Conductors.
ASTM-B 258 - Standard Nominal Diameters and Cross-Sectional Areas of AWG Sizes of Solid Round Wires Used As Electrical Conductors.
ASTM-B 286 - Copper Conductors For Use In Hookup Wire For Electronic Equipment.
ASTM-B 355 - Nickel-Coated Soft or Annealed Copper Wire.
ASTM-D 297 - Rubber Products, Chemical Analysis of.
ASTM-D 1458 - Fully Cured Silicone Rubber-Coated Glass Fabric and Tapes for Electrical Insulation, Testing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNDERWRITERS' LABORATORIES, INC.

(Application for copies of publications should be addressed to Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, New York 11746 or 207 East Ohio Street, Chicago, Illinois 60611.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.
C96.1 - Temperature Measurement Thermocouples

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
Insulated Power Cable Engineers Association (IPCEA)
Publication No. S-66-524 - Cross-linked-thermosetting-polyethylene-insulated Wire and Cable for the Transmission and Distribution of Electrical Energy (NEMA Publication No. WC 7-1971).

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Detail requirements. Detail requirements or exceptions to the general requirements specified herein shall be as specified by the specification sheet. In the event of any conflict between the general requirements of this specification and the specification sheet, the latter shall govern. Except for those cables having a silicone rubber jacket, the preferred construction of all cables is without armor. Cables for nonflexing service and for which the preferred construction is without armor, may be procured with armor. The special requirements for such alternate construction are shown on the applicable specification sheets.

3.2 Qualification. Cables and cords furnished under this specification which require qualification (see applicable specification sheet) shall be products which are qualified for listing on QPL-915 at the time set for opening of bids (see 4.3 and 6.5).

3.2.1 Requalification. Changes in materials and construction shall require the written approval of the Naval Ship Engineering Center (NAVSEC) Code 6158F. Incorporation of any changes which have not been so approved shall require requalification of the cable or cord or groups of cables or cords in question.

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3.3 Materials. All materials used in the construction of those cables or cords requiring qualification shall have the written approval of NAVSEC. In the case of cables or cords not requiring qualification, NAVSEC approval will not be required, and all materials in these cables or cords shall conform to the requirements shown in the applicable specification sheets.

3.3.1 Conductors.

3.3.1.1 Standard copper conductors. Standard copper conductors shall be composed of soft or annealed copper strands conforming to ASTM-B3.

3.3.1.2 High strength conductors. Unless otherwise specified on the specification sheet high strength conductors shall be composed of 30 percent conductivity, high strength, copper-covered steel strands conforming to ASTM-B228.

3.3.1.3 Tin coating. Tin coating shall be commercially pure tin conforming to ASTM-B33.

3.3.1.4 Lead or lead-alloy coating. Lead or lead alloy coating shall conform to ASTM-B189.

3.3.1.5 Nickel coating. Nickel coating shall conform to Class 2 of ASTM-B355.

3.3.1.6 Thermocouple wire. Conductors for thermocouples shall conform to ANSI Standard C96.1.

3.3.2 Insulation. The silicone rubber-coated glass fabric tape insulation shall, before application to a conductor, have a dielectric strength of not less than 475 volts per mil for tapes which are less than 10 mils thick, and 575 volts per mil for tapes which are 10 mils or more in thickness. These dielectric strength values shall be met after the tapes have been conditioned for 24 hours at 23°C and 96 percent relative humidity. The test procedure shall be the short-time dielectric breakdown test of ASTM-D 1458. Extruded silicone rubber insulation shall have a specific gravity of not more than 1.55. Other physical and electrical properties shall be as specified by the specification sheets.

3.3.3 Jackets. The material used for jackets over insulated conductors and groups of insulated conductors, and for the cable or cord jacket, shall have the physical and electrical properties as required by the specification sheet. Unless otherwise specified on the specification sheet all thermoplastic cable or cord jackets shall be gray - color No. 26270 of FED-STD-595.

3.3.3.1 Thermoplastic jackets . Thermoplastic jackets are polyvinyl chloride and are classified as standard thermoplastic or special thermoplastic. The special thermoplastic jacket is used where necessary to provide extra protection. The standard thermoplastic jacket is used for lightweight cables or cords which are classified for either flexing or non-flexing service and the jacket shall be compounded to meet the requirements of the specification sheet. When a specification sheet specifies a special thermoplastic jacket, a standard thermoplastic shall be used if the cable is procured with armor.

3.3.4 Shields. The materials and constructions for shields of insulated conductors, groups of insulated conductors and overall cable or cord shall be as required by the specification sheet. When AWG sizes are specified for wire shields, they shall be in accordance with ASTM-B258.

3.3.5 Fillers. Unless otherwise specified on the applicable specification sheet, fillers for cables or cords which are not required to be watertight may be fibrous. Fillers for cables or cords which are required to be watertight shall be non-fibrous or a combination of non-fibrous and fibrous as required by the specification sheet.

3.3.5.1 Fibrous fillers. Fibrous fillers shall consist of either cotton, jute, synthetic fibers, glass fibers or asbestos. The material selected shall be appropriate for the type of cable or cord in which it is used, and shall have been treated for flame or moisture resistance, or a combination of both, as necessary, to enable the cable or cord to meet the requirements for the particular type of cable or cord.

3.3.5.2 Non-fibrous fillers . Non-fibrous fillers shall consist of elastomeric material which is readily removable from insulation of conductors and insulating tapes over shields without the aid of solvents, cleansers or tools. The acceptability of the material shall be determined by the cable filler removability test and the applicable cable aging and compatibility test.

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3.3.6 Tapes. Tapes shall be of the material as specified in the specification sheet or shall be a type approved by NAVSEC for the specific cable or cord construction.

3.3.7 Braids (identification). Colored braids used for conductor identification shall be of rayon in accordance with MIL-C-572.

3.3.8 Braids (glass). Glass braids for use on silicone insulated conductors shall be composed of the appropriate size of either staple or continuous fiber conforming to MIL-Y-1140.

3.3.9 Tie cords. Tie cords shall be of cotton, glass, or synthetic fiber having a maximum diameter of 0.065 inch and a minimum breaking strength of 30 pounds. When necessary tie cords shall be treated with a flexible waterproofing compound.

3.3.10 Separators. Unless otherwise specified in the specification sheet, separators, when required, shall consist of a wind of glass fibers, synthetic fibers, or cotton; they may also be of a tape of cotton, synthetic fiber, paper or polyester. All separators shall be opaque and the choice of separator material shall be appropriate for the type of cable or cord in which is is employed.

3.3.11 Reinforcement. Reinforcing binders or reinforcement for a single or double layer jacket shall be a size 3 nylon seine twine having a minimum breaking strength of 15 pounds. Alternate reinforcement may be used, only when approved by NAVSEC.

3.3.12 Binders. Binders shall be of the material specified in the specification sheet, or shall be of a material approved by NAVSEC for the specific cable or cord construction.

3.3.13 Armor wires. The armor wires shall consist of Alclad 5056 aluminum alloy having a diameter of 0.0126 ±0.0005 inch, a minimum tensile strength of 50,000 pounds per square inch (psi) and a minimum elongation (before application to the cable) of 2 percent in 10-inch length.

3.3.14 Aluminum paint. Aluminum paint shall consist of aluminum pigment (paste form) conforming to TT-P-320, in a suitable vehicle of the synthetic resinuous (Phenol-formaldehyde or glyceride) type designed for good penetration, high bending power, resistance to water and oil, and with flexibility at temperatures up to 100°C.

3.4 Construction.

3.4.1 Conductor stranding. The size and quantity of individual conductor strands and the total circular-mil area of each conductor shall be in accordance with table I, II or III as applicable in accordance with the size designation specified by the specification sheet.

3.4.1.1 Concentric-lay-stranded. The length and direction of lay and the type and number of joints in concentric-lay-stranded conductors shall be in accordance with ASTM-B8 or ASTM-B286, as applicable.

3.4.1.2 Bunch-stranded. The length and direction of lay and the type and number of joints in bunch-stranded conductors shall be in accordance with ASTM-B174.

3.4.1.3 Rope-lay-stranded. The length and direction of lay and the type and number of joints in rope-lay-stranded conductors shall be in accordance with ASTM-B172 or ASTM-B173, as applicable.

Table I - Conductor data, concentric-lay stranded (ASTM B-286).

Conductor Size ASTM-B286 designation	Number of Strands (minimum)	Strand Diameter, Nominal (inch)	Conductor Diameter, Nominal (inch)	Conductor cross-sectional area, nominal (circular mils)	Weight per 1000 feet approximate (pounds)
12-37	37	0.0126	0.088	5874	20.20
12-19	19	.0179	.091	6088	20.20
14-19	19	.0142	.072	3831	12.65
16-19	19	.0113	.057	2426	7.97
18-19	19	.0100	.051	1900	5.02
18-7	7	.0159	.049	1770	5.02

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Table I - Conductor data, concentric-lay stranded (ASTM B 286) (Cont'd).

Conductor Size ASTM-B286 designation	Number of Strands (minimum)	Strand Diameter, Nominal (inch)	Conductor Diameter, Nominal (inch)	Conductor cross-sectional area, nominal (circular mils)	Weight per 1000 feet approximate (pounds)
20-19	19	.0080	.041	1216	3.16
20-7	7	.0126	.038	1111	3.16
22-19	19	.0063	.032	754	1.98
22-7	7	.0100	.031	700	1.98
24-19	19	.0050	.026	475	1.24
24-7	7	.0080	.025	448	1.24
26-19	19	.0040	.021	304	.780
26-7	7	.0063	.020	278	.780
28-19	19	.0031	.016	183	.490
28-7	7	.0050	.016	175	.490
30-7	7	.0040	.013	112	.309

Table II - Conductor data, concentric-lay stranded (Navy Standard).

Conductor size Navy Standard	Number of strands (Minimum)	Strand diameter Nominal (inch)	Conductor diameter Nominal (inches)	Conductor cross-sectional area (circular mils)		Max. conductor resist. (d.c.) per 1000 feet at 25°C.[1/]		Weight per 1000 feet Approx. (Pounds)
				(Nominal)	(Minimum)	Bare (Ohms)	Coated (Ohms)	
1(7)	7	0.013	0.038	1,119	1,096	9.89	10.3	3.4
2(7)	7	.016	.048	1,779	1,743	6.25	6.50	5.5
3(7)	7	.020	.060	2,828	2,771	3.92	4.07	8.7
3(19)	19	.013	.063	3,036	2,975	3.64	3.80	9.3
4(7)	7	0.025	0.076	4,497	4,407	2.46	2.52	14
6(7)	7	.031	.092	6,512	6,382	1.69	1.73	20
9(7)	7	.036	.108	9,016	8,836	1.23	1.25	28
14(7)	7	.045	.136	14,340	14,050	0.770	0.784	44
23(7)	7	.057	.171	22,800	22,340	.486	.493	70
30(19)	19	.040	.202	30,860	30,240	.358	.365	95
40(19)	19	.045	.226	38,910	38,130	.284	.288	120
50(19)	19	.051	.254	49,080	48,100	.225	.228	150
60(37)	37	.040	.282	60,090	58,890	.185	.189	190
75(37)	37	.045	.317	75,780	74,260	.146	.149	230
100(61)	61	.040	.363	99,060	97,080	.112	.115	310
125(61)	61	.045	.407	124,900	122,400	.0888	.0904	390
150(61)	61	.051	.457	157,600	154,400	.0704	.0716	490
200(61)	61	.057	.514	198,700	194,700	.0560	.0570	610
250(61)	61	.064	.577	250,500	245,500	.0444	.0453	770
300(91)	91	.057	.628	296,400	290,500	.0375	.0382	910
350(91)	91	.062	.682	349,800	342,800	.0316	.0321	1,100
400(127)	127	.057	.742	413,600	405,400	.0268	.0273	1,300
500(127)	127	.064	.832	521,600	511,100	.0214	.0217	1,600
650(127)	127	.072	.936	657,600	644,500	.0169	.0172	2,000
800(127)	127	.081	1.050	829,300	812,700	.0134	.0136	2,600
1000(127)	127	.091	1.180	1,046,000	1,025,000	.0106	.0108	3,200
1300(127)	127	.102	1.325	1,318,000	1,292,000	.00843	.00851	4,100
1600(127)	127	.114	1.485	1,662,000	1,629,000	.00668	.00676	5,100
2000(127)	127	.128	1.670	2,097,000	2,055,000	.00530	.00536	6,300

[1/] Coated refers to tin or lead-tin alloy.

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Table III - Conductor data, bunch and rope-lay stranded (Navy Standard)

Conductor size, Navy Standard	Number of strands Minimum	Strand diameter Nominal (Inch)	Conductor Diameter Nominal (Inches)	Conductor cross-sectional area (Circular mils)		Max. conductor resist. (d.c.) per 1000 feet at 25°C. 2/		Weight per 1000 feet Approx. (Pounds)
				Nominal (Cir. Mils)	Minimum (Cir.mils)	Bare (Ohms)	Coated (Ohms)	
Bunch:								
1/2(21)	21	0.005	0.028	525	515	21.1	22.2	1.6
3/5(7)	7	.010	.030	703	689	15.7	16.4	2.2
1(10)	10	.010	.038	1,005	985	11.0	11.5	3.1
1(26)	26	.006	.042	1,034	1,014	10.7	11.2	3.2
1-1/2(16)	16	.010	.049	1,608	1,576	6.88	7.17	4.9
1-1/2(41)	41	.006	.049	1,630	1,597	6.78	7.15	5.0
2-1/2(19)	19	.011	.057	2,426	2,300	4.75	4.90	6.3
2-1/2(26)	26	.010	.061	2,613	2,561	4.23	4.48	8.0
2-1/2(65)	65	.006	.061	2,594	2,542	4.26	4.51	8.0
4(41)	41	.010	.077	4,121	4,038	2.68	2.81	13
6(65)	65	.010	.097	6,533	6,402	1.69	1.78	20
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope:								
20(49)	7x7	.020	.180	19,800	19,400	.562	.582	61
23(228)	19x12	.010	.190	22,910	22,460	.499	.523	73
26(49)	7x7	.023	.210	26,250	25,730	.428	.440	82
30(304)	19x16	.010	.220	30,550	29,940	.374	.392	97
33(336)	7x48	.010	.235	33,370	33,090	.344	.360	106
42(209)	19x11	.014	.260	42,110	41,280	.272	.284	130
42(49)	7x7	.029	.260	41,740	40,910	.268	.275	130
53(532)	19x28	.010	.304	53,470	52,400	.213	.218	169
60(304)	19x16	.014	.310	61,260	60,040	.187	.196	190
66(133)	19x7	.022	.330	66,370	65,040	.171	.175	210
84(2107)	2107	1/	.410	83,690	82,020	.138	-----	270
83(418)	19x22	.014	.380	84,230	82,560	.136	.142	270
105(2646)	2646	1/	.460	105,500	103,400	.108	-----	340
105(259)	37x7	1/	.410	105,500	103,400	.108	.113	330
133(3325)	3325	1/	.520	133,100	130,400	.0876	-----	430
133(259)	37x7	1/	.460	133,100	130,400	.0856	.0892	420
133(684)	19x36	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	19x40	.014	.510	153,100	150,100	.0747	.0780	490
168(427)	61x7	1/	.520	167,800	164,400	.0681	.0710	530
200(988)	19x52	.014	.580	199,100	195,100	.0575	.0600	630
220(259)	37x7	.029	.610	220,700	216,300	.0494	.0507	718
250(1254)	19x66	.014	.680	252,700	247,700	.0453	.0472	800
250(6384)	19x7x48	.006	.713	250,000	245,000	.0461	------	821
300(259)	37x7	.034	.714	300,000	294,000	.0378	------	926
400(2052)	19x108	.014	.850	413,500	405,300	.0277	.0289	1,300
500(259)	37x7	.044	.922	500,000	490,000	.0234	-----	1,585
600(427)	61x7	.038	1.013	600,000	588,000	.0196	-----	1,910
650(427)	61x7	.039	1.053	650,000	637,000	.0154	-----	2,070
800(4033)	37x109	.014	1.150	812,700	796,500	.0141	.0148	2,600

1/ These are conductors for flexible cables, for which the minimum number of strands and minimum area are specified. The individual strand diameter is determined from the required area and the actual number of strands used.

2/ Coated refers to tin or lead-tin alloy.

3.4.2 Separators. Separators employed directly over conductors shall be applied so as to give 100-percent coverage to the conductors.

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3.4.3 Insulation.

3.4.3.1 Extruded insulation. Extruded insulations shall be applied concentrically and to the dimensions required by the specification sheet.

3.4.3.2 Taped insulation. Tapes used as insulation shall be applied in such a manner that they lie smoothly and free from wrinkles. The tape width shall be proper for the diameter over which it is applied so as to minimize splits, creases and edge tears when the completed cable is subjected to bending. Registrations shall be held to a minimum, consistent with the best commercial practice.

3.4.3.2.1 Silicone rubber-glass tapes. Silicone rubber-glass tapes shall be held from unwinding from conductors when the conductors are exposed and subjected to normal handling during splicing and terminating. The tapes may be held from unwinding by the use of self-bonding tapes, by the use of a silicone rubber adhesive or by using self-bonding or heat sealable polyester tapes over the insulating tapes. All voids within the insulating wall shall be filled with a compound so as to exclude air and moisture and prevent the passage of water through the insulation. Any adhesive or sealing compound shall have no deleterious effect on the other cable parts during tests or during the normal service life of the cable.

3.4.4 Glass braids. Glass braids for use on extruded silicone rubber insulation shall have a thickness of approximately 6 mils. The braid angle shall be such that the completed cable will conform to all inspection and test requirements.

3.4.5 Braid covering (over glass braid). When a specification sheet requires a covering over the glass braid, the covering shall consist of a layer or layers of a black material that will provide high resistance to moisture, abrasion and hydraulic fluids and also provide a smooth surface for printed identification marking.

3.4.6 Fillers. Fillers shall be used to provide firmness and roundness of completed cables or cords and to provide watertightness when required. The type of filler material shall be as specified in 3.3.5.

3.4.7 Binder. A binder shall be employed over the assembled core of all cables or cords to provide a firm core and to serve as a barrier, when necessary, between the core and cable or cord jacket.

3.4.8 Shields. The material and construction of the shields shall be as specified in the specification sheet.

3.4.8.1 Braided wire shield. Braided wire shields shall have the angle of application and the percent coverage as required in the specification sheet. The percent coverage and the angle of application shall be determined by the following formula:

$$\text{Percent coverage } (k) = 100(2F-F^2)$$

Where $F = NPd/\sin a$

$\tan a = 2\pi DP/C$

a = acute angle of braid with axis of cable or cord
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires per carrier
C = number of carriers
P = picks per inch of cable or cord length

3.4.8.2 Served wire shield. When served wire shield is required, it shall be composed of a serve or wrap of fine copper wires evenly spaced and providing the coverage as specified in the specification sheet. The percent coverage shall be determined by the following formula:

$$\text{Percent coverage} = \frac{Nd}{W} \times 100$$

Where N = Number of parallel wires
d = Diameter of individual wires in inches
$W = \pi D \cos a$
D = Diameter under shield in inches
a = Angle between serving wires and axis of cable or wire
$\tan a = \pi \frac{D}{C}$
C = Pitch of serving in inches

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3.4.9 Identification codes and methods. Individual conductors and groups of conductors shall be separately identified. The applicable identification code and the method by which the code is applied shall be as specified in the specification sheet.

3.4.9.1 Identification codes.

3.4.9.1.1 Standard identification code. Standard identification code shall be in accordance with table IV.

Table IV - Standard Identification code.

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
1	Black	---	---
2	White	---	---
3	Red	---	---
4	Green	---	---
5	Orange	---	---
6	Blue	---	---
7	White	Black	---
8	Red	Black	---
9	Green	Black	---
10	Orange	Black	---
11	Blue	Black	---
12	Black	White	---
13	Red	White	---
14	Green	White	---
15	Blue	White	---
16	Black	Red	---
17	White	Red	---
18	Orange	Red	---
19	Blue	Red	---
20	Red	Green	---
21	Orange	Green	---
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	---	---
39	Brown	Black	---
40	Brown	White	---
41	Brown	Red	---
42	Brown	Green	---
43	Brown	Orange	---
44	Brown	Blue	---
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red
48	Orange	Red	Blue
49	Blue	Red	Orange

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Table IV - Standard Identification Code (Cont'd).

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange
81	Blue	Black	Green
82	Orange	White	---
83	Green	Red	---
84	Black	Green	---
85	White	Green	---
86	Blue	Green	---
87	Black	Orange	---
88	White	Orange	---
89	Red	Orange	---
90	Green	Orange	---
91	Blue	Orange	---
92	Black	Blue	---
93	White	Blue	---
94	Red	Blue	---
95	Green	Blue	---
96	Orange	Blue	---
97	Yellow	---	---
98	Yellow	Black	---
99	Yellow	White	---
100	Yellow	Red	---
101	Yellow	Green	---
102	Yellow	Orange	---
103	Yellow	Blue	---
104	Black	Yellow	---
105	White	Yellow	---
106	Red	Yellow	---
107	Green	Yellow	---
108	Orange	Yellow	---
109	Blue	Yellow	---
110	Black	Yellow	Red

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Table IV - Standard Identification Code (Cont'd).

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
111	White	Yellow	Red
112	Green	Yellow	Red
113	Orange	Yellow	Red
114	Blue	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.4.9.1.2 Telephone identification code. Conductor identification code for telephone cables shall be as follows:

Color or Conductor No.	Color	Color or Conductor No.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

The mating of conductors for forming pairs shall be as follows:

No. 1 mated with Nos. 2 thru 12 for first 11 pairs
No. 2 mated with Nos. 3 thru 12 for next 10 pairs
No. 3 mated with Nos. 4 thru 12 for next 9 pairs
No. 4 mated with Nos. 5 thru 12 for next 8 pairs
No. 5 mated with Nos. 6 thru 12 for next 7 pairs
No. 6 mated with Nos. 7 thru 12 for next 6 pairs
No. 7 mated with Nos. 8 thru 12 for next 5 pairs
No. 8 mated with Nos. 9 thru 12 for next 4 pairs
No. 9 mated with Nos. 10 thru 12 for next 3 pairs
No. 10 mated with Nos. 11 thru 12 for next 2 pairs
No. 11 mated with No. 12

3.4.9.1.3 Special identification code. Special identification code shall be in accordance with the following:

Color or Conductor No.	Color	Color or Conductor No.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

3.4.9.1.4 Letter identification code. Letter identification code shall consist of the letters A, B, C, and D printed in block type and with black, white, red, and green ink, respectively.

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3.4.9.2 Identification methods.

3.4.9.2.1 Method 1. Identification method 1 shall be surface printing of both number and color designations. The legend shall be printed in contrasting color: preferably white ink on black or dark background or black ink on white or light background. The legend shall be repeated at intervals not exceeding 3-inches and alternate legends shall be inverted. For example: 10-ORANGE-BLACK 10-ORANGE-BLACK The character type shall be block or italic and shall have a height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character Approx. (inch)
0.045 to 0.070	0.025
.070 to .095	1/32
.095 to .115	3/64
.115 to .200	1/16
.190 to .250	5/64
.235 to .375	3/32
.330 and larger	1/8

3.4.9.2.2 Method 2. Identification method 2 shall be the use of opaque white polyester tapes which have been printed with both the number and color designation prior to application. The legend shall be printed with black ink and shall be repeated at intervals not exceeding 3 inches. The character type shall be block and shall be approximately 3/32 inch high.

3.4.9.2.3 Method 3. Identification method 3 shall be the use of solid base colors or solid base colors with tracers as required. The base color may be either the color of the insulation or the color of a coating applied to the insulation. The tracers shall be approximately 1/32-inch wide ink stripes of the required color applied helically with 1-1/2 ±1/4 inch lay.

3.4.9.2.4 Method 4. Identification method 4 shall be the use of colored braids. Tracers shall consist of the required colors applied by three adjacent carriers. Where two tracers are required, they shall be applied with reverse lay.

3.4.9.2.5 Method 5. Identification method 5 shall be the use of the printed letter on the outermost insulating tape or the printed letter on a polyester binder tape over the insulating tapes. The letters shall be approximately 3/16 inch high and shall have been printed at intervals not exceeding 3 inches prior to the application of the tape to the conductor.

Note: If the insulating tapes are white, no printing is required on the B (white) conductor.

3.4.10 Manufacturer's identification tape. Unless otherwise indicated on the specification sheet, all cables and cords shall contain a continuous, thin, moisture-resistant marker tape, not less than 1/10 inch wide. The marker tape shall be placed directly under the cable or cord binder tape or jacket unless otherwise approved by NAVSEC. The tape shall be printed to show the following information at intervals not greater than 1 foot: Name and location of manufacturer; year of manufacture; specification number (MIL-C-915); and progressive serial number. The serial number is not necessarily a footage marker. A serial number shall not be repeated by a manufacturer in any one year for any one type and size of cable or cord.

3.4.10.1 Marker threads. The use of marker threads in lieu of the manufacturer's identification tape, in small cables or cords, shall have written approval of NAVSEC. If approved, marker threads shall be in accordance with those assigned by Underwriter's Laboratory (U.L.) Incorporated, to indicate the manufacturer of the cable or cord. When U.L. marker threads are used, yearly marker threads shall also be used to indicate the year of manufacture, as follows:

Year	Color Codes
1971	Red-Yellow
1972	Red-Slate
1973	Red-Green
1974	Red-Brown
1975	Red-Black
1976	Red-Pink

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Year	Color Codes
1977	Blue-Yellow
1978	Blue-Slate
1979	Blue-Green
1980	Blue-Brown
1981	Blue-Black

3.4.11 Cable or surface marking. When overall cable or cord jacket surface marking is required by the specification sheet, ink marking shall be used. The legend shall be printed in contrasting color: preferably white ink on a dark background or black ink on a light background. The legend shall consist of the manufacturer's name, the cable or cord type and size[1/] designation, the type of jacket material and the year of manufacture. The legend shall be repeated at intervals not exceeding 1 foot and the year of manufacture need not be in line with the balance of the legend.

For example: "Manufacturer's name - MWF-24-NEOP - 1973"

The character type may be either block or italic and shall have a minimum height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character Approx. (inch)
0.125 to 0.200	3/64
.200 to .285	1/16
.285 to .350	5/64
.350 to .500	3/32
.500 and larger	1/8

Jacket material names shall be abbreviated as follows:

Polychloroprene: NEOP
Standard thermoplastic: STD PVC
Special Thermoplastic: SPL PVC
Chlorosulfonated polyethylene: CSPE

3.4.12 Watertightness. When watertight cable construction is specified in the specification sheet, all voids within the cable or cord construction shall be filled to prevent the passage of water longitudinally through the cable or cord.

3.4.13 Jacket reinforcement. When a reinforced polychloroprene jacket is required by the specification sheet, the jacket shall be applied in two layers with a size 3 nylon cord applied as a braid or two reverse serves between the layers. The minimum thickness of the outer layer of jacket shall be 50 percent of the total. The reinforcement shall be applied with approximately four picks or crossovers per inch.

3.4.14 Braided armor. The armor shall be applied in the form of a braid. The weave of the braid shall be of either the "one over-one under" or the "two over-two under" type. The weave shall be such that the wires of a carrier are laid closely together, flat and parallel. The braid shall be applied with maximum tension possible so as to prevent loosening and creeping but not to cause broken ends. The braid shall remain snug to the cable jacket and not spring away when cut. Splices in the braid strands shall be infrequent, staggered, and inconspicuous so as not to increase the nominal diameter of the cable or result in rough spots. The selection of the number of wires per carrier and the number of carriers per braider shall be such as to produce a basket weave to give a minimum of 88 percent coverage and a braid angle within the limits shown in table V. The number of wires per carrier shall be not more than shown in table VI.

#

Table V - Armor braid angle.

Diameter over jacket (Inches)	Braid angle	
	Minimum (Degrees)	Maximum (Degrees)
0.100 thru 0.600	30	60
.601 thru 1.000	35	60
1.001 thru 1.500	40	70
1.501 thru 2.000	45	70
2.001 thru 3.000	50	80

1/ When applicable.

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Table VI - Wires per carrier.

Diameter under braid (Inches)	Maximum number of ends per carrier	
	One over-one under	Two over-two under
0.100 thru 0.400	8	5
.401 thru .800	12	10
.801 thru 3.000	16	10

3.4.14.1 Coverage. The 88 percent minimum coverage shall be as determined by the following formula:

$$\text{Percent coverage } (K) = 100(2F-F^2)$$

Where, F = NPd/Sin a
Tan a = 2 // DP/C
a = acute angle of braid with axis of cable
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires in carrier
C = number of carriers
P = picks per inch of cable length

3.4.15 Armor paint. Aluminum paint shall be applied to the armor as soon as practicable after completion of manufacturing operations so as to prevent accumulation of dirt, moisture, and foreign matter on the braid or in the interstices. The paint shall be properly dried and shall not be tacky. The painting shall be done preferably by the immersion method so as to fill all interstices to the outer cable surface. The paint finish shall provide a relatively smooth overall coating that will facilitate making watertight glands with stuffing tubes and packing. The coating shall be flexible and adhere firmly so that it will not crack, peel, or flake off.

3.4.16 Dimensional tolerances. Where minimum or maximum dimensions, or both, are specified, no minus or plus tolerances, respectively, will be permitted. Where a dimension is specified as nominal, the average dimension shall be not less than the specified nominal. Where no minimum overall cable or cord diameter is specified, the minimum permissible diameter shall be not less than 92-1/2 percent of the specified maximum overall cable or cord diameter. Where a specification sheet gives the maximum overall diameter for unarmored cables and the cable is procured with armor, the maximum overall diameter shall be increased by 0.050 inch.

3.4.16.1 Conductor insulation wall thickness. For conductor insulation wall thickness specified as nominal, the average thickness shall be not less than the specified nominal. The minimum thickness, measured at any cross section, shall be not less than 90 percent of the specified nominal.

3.4.16.2 Cable or cord jacket thickness. The average thickness of a cable or cord jacket measured at any cross section shall be not less than the specified nominal. In case of multiconductor cables or cords the jacket thickness shall be determined from the measurements made at the high point of each conductor taken on a line through the center of the cable or cord and through the center of each conductor in the outer layer. The minimum thickness at any cross section shall be not less than 80 percent of the specified nominal.

3.4.17 Centering and circularity.

3.4.17.1 Insulation. The insulation on the individual conductors shall be uniform in diameter throughout the conductor length. At any cross section, the maximum wall thickness shall not exceed the minimum by more than 25 percent for specified thickness greater than 0.025 inch, nor by more than 40 percent for specified thicknesses of 0.025 inch and less.

3.4.17.2 Cable or cord jacket. The cable or cord jacket shall be applied concentrically to the cable or cord core in a manner to maintain circularity in the completed cable or cord. The maximum wall thickness of the jacket at any cross section shall not exceed the minimum by more than 66 percent.

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3.4.18 Surface condition (cable or cord jacket). The surface of the cable or cord jacket of all unarmored cables or cords shall be dry and free from any coating, film or treatment which would tend to interfere with the bonding to it of encapsulating or molding materials normally used in splicing and terminating.

3.5 Electrical properties. All electrical properties of the completed cable or cord shall be as required by the specification sheet.

3.6 Physical properties. All physical properties of the completed cable or cord and cable or cord components shall be as required by the specification sheet.

3.7 Repair of insulation or cable or cord jacket. Repair of cable or cord jacket will not be permitted on unarmored cables which are required to pass a hydrostatic test. Repair of cable or cord jacket will be permitted on cables or cords which are not required to pass a hydrostatic test, provided the materials and techniques used are such that the finished cable or cord complies with all the requirements of this specification. The materials and techniques used in the repair of either insulation or cable or cord jacket shall be subject to approval of NAVSEC. The frequency of repairs shall be held to a minimum.

3.8 End seals. All cables and cords covered by this specification shall have both ends of each shipping length sealed to prevent the entrance of moisture. Materials used for end seals and method of application shall have the written approval of NAVSEC.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performing (or having performed) all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities (approved by NAVSEC) for performing (or having performed) the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of all cable, and cord covered by this specification shall be classified as follows:

(a) Qualification inspection
(b) Specification conformance inspection
    (1) Basic electrical
    (2) Group A examination and test
    (3) Group B tests
    (4) Group C tests
(c) Comparison inspection
(d) Conductor inspection

4.3 Qualification tests.[2/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examination and tests specified on the applicable specification sheet.

4.3.1 Qualification specimens. The manufacturers seeking qualified products listing shall manufacture and be responsible for testing at least one specimen of cable or cord of each type, size or group for which qualification is sought. The cable or cord type and size required for group qualification shall be in accordance with table VII. When a manufacturer desires qualification of an individual type or size, the selection of test specimens shall be subject to review by NAVSEC. A sufficient length of any test specimen must be manufactured at one time. Suggested manufacturing lengths for test specimens are given in table VII. The groups are based on similar characteristics and requirements. The Naval Ship Engineering Center may extend approval of any one type and size of cable or cord to cover any or all types in the same group; however, NAVSEC may require performance of any or all qualification inspections on additional types in the same group prior to such extension of approval.

2/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5 through 6.5.2).

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Table VII - Qualification specimens and groups.

Group No.	Qualifying Type and Size 1/	Types and Sizes Comprising the Group
1	DLT	DLT
2	DRW	SRW, DRW, TRW
3	DSS-3	All sizes of DSS, TSS, FSS, 5SS and 7SS
4	DSWS	DSWS
5	ECM	ECM, MSP and MSPW.
6	JAS	JAS
7	MCOS-6	All sizes of MCOS
8	MCSF	MCSF
9	MDU-23	All sizes of MDU
10	MHOF-30	All sizes of MHOF
11	MMOP	All sizes of DCOP, TCOP and MMOP
12	MNW-10	All sizes of MNW
13	MSCU-10	All those sizes of DSGU, TSGU, FSGU, MSCU, 7SGU, TCJU AND TCTU which have conductor sizes 23 and smaller.
14	MWF-24	All sizes of MWF
15	PBTMU-15	All sizes of PBTMU and TSP
16	PI-12	All sizes of PI, DPS, TPS, FPS and 7PS
17	S2S	S2S
18	TCTX-7	All sizes of TCJX, TCKX AND TCTX.
19	THOF-9	Sizes 42 and smaller of types DHOF, THOF AND FHOF, excepting type-and-size THOF-14.
20	THOF-150	Sizes 60 and larger of types DHOF, THOF and FHOF; type-and-size THOF-14; type CVSF.
21	TNW-150	All sizes of DNW, TNW and FNW.
22	TPNW-20	All sizes of TPNW.
23	TPU	TPU
24	TSGU-150	All those sizes of SSGU, DSGU, TSGU, FSGU and 6SGU which have conductor sizes 50 and larger.
25	TTOP-10	All sizes of TTOP
26	TTRS-8	All sizes of TTRS
27	TTSU-10	All sizes of TTSU
28	1SU-36	All sizes of 1SU
29	1S50MU-16	All sizes of 1SAU, 1SMU, 1S50MU and 1S75MU.
30	2AU-40	All sizes of MS, MU, 2AU, 2SJ, 2U, 3SJ and 4SJ.
31	2SWAU-10	All sizes of 1SMWU, 1SWU, 2SWAU, 2SWU, 2SU, 2WAU, 3SU, 3SWU and 3U.
32	2SWF-7	All sizes of 1SWF, 2SWF and 3SF.
33	5KVTSGU-100	All sizes of 5KVTSGU.

1/ Suggested specimen length is 200 feet for all qualifying types and sizes, except 2SWAU-10, for which it is 500 feet.

4.4 Definition of terms. For the purpose of this section, the following definitions apply:

4.4.1 Inspection lot. An inspection lot shall consist of the total number of units of product of any one type, size, and construction manufactured under the same conditions and offered for delivery at one time.

4.4.2 Unit of product. A unit of product is the unit ordering length as specified on the specification sheet, except that when the unit ordering length is 500 feet, two such lengths shall constitute a unit of product. Random and remnant lengths or special order lengths may be added together to equal a unit ordering length.

Note: Length tolerances as given for standard lengths shall apply. (See 6.2.2).

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4.4.3 Sample. A sample is one unit of product selected from an inspection lot.

4.4.4 Specimen. A specimen is an individual length of cable or cord or an individual length of a part of cable or cord which has been taken from a sample.

4.4.5 Defective unit. A defective unit is a unit of product from which a specimen that failed to meet one or more requirements was taken.

4.4.6 Rejected lot. A rejected lot is a lot which contains more than the acceptable number of defective units.

4.5 Specification conformance inspection. Specification conformance inspection shall be performed on all completed cable or cord in accordance with the procedures given or referenced herein. This inspection shall consist of basic electrical tests plus groups A, B and C examination and tests in accordance with the specification sheets.

4.5.1 Basic electrical tests. Basic electrical tests shall consist of voltage withstand, insulation resistance and conductor resistance. These tests shall be performed on each length of completed cable or cord. The conductor resistance test shall be performed on all conductors which have a nominal circular mil area of less than 1600; for conductors which have a nominal circular mil area of 1600 or greater, the conductor resistance test shall be performed on conductors selected as follows:

No. of insulated conductors in cable or cord	No. of insulated conductors selected for test
1 thru 4	All
5 thru 20	4
21 thru 30	5
31 thru 42	6
43 thru 56	7
57 thru 72	8
73 thru 91	9
92 thru 110	10
111 thru 132	11
133 thru 156	12

4.5.2 Sampling procedure. The required number of samples for groups A, B and C examination and tests shall be selected at random from the inspection lot. All nonconforming starting and finishing ends of completed cable or cord shall be removed by the manufacturer prior to selecting samples.

4.5.2.1 Sampling for group A examination and tests. Samples for group A examination and tests shall be selected from each lot in accordance with the first two columns of table VIII.

Table VIII - Sampling for group A examination and tests.

Units of product in lot	Number of samples	Acceptance number defective units	Rejection number defective units
10 or under	2	0	1
11 thru 30	3	0	1
31 thru 90	7	0	1
91 thru 210	15	1	2
211 and over	25	2	3

4.5.2.2 Sampling for group B tests. Samples for group B tests shall be selected from each lot in accordance with table IX.

Table IX - Sampling for group B tests.

Units of products in lot	Number of samples
8 and under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

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Note 1: The number of conductors from which specimens of insulation shall be taken shall be chosen as follows:

Number of insulated conductors in sample	Number of insulated conductors to be chosen for specimens
1 thru 4	All
5 thru 20	4
21 thru 30	5
31 thru 42	6
43 thru 56	7
57 thru 72	8
73 thru 91	9
92 thru 110	10
111 thru 132	11
133 thru 156	12

Note 2: At least one specimen of insulation shall be tested from each conductor chosen. The minimum number of specimens of insulation and jacket shall be five. The test result shall be the median value obtained.

4.5.2.3 Sampling for group C tests. Samples for group C tests shall be selected in accordance with table X.

Table X - Sampling for group C tests.

Two months' production (units of product)	Number of samples
8 or under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

4.5.2.4 Tightened sampling and accept-reject criteria. If the number of defective units in a lot exceeds the acceptance number for group A examination and tests (see table VIII), or if there are any failures in group B tests, the manufacturer shall have the option of reworking the lot and performing examination and tests on the tightened sampling basis in accordance with table XI. If there are any failures on tests performed on the tightened sampling basis, if there are any failures in group C tests, or if the manufacturer chooses not to rework and retest, the lot shall be rejected.

Table XI - Tightened sampling.

Units of product in lot	Number of samples
10 or under	3
11 thru 30	5
31 thru 90	9
91 thru 210	15

4.5.2.5 Action in case of rejection. Rejected cable, or cord may be referred to the Government for determination of acceptability of the point or points of non-compliance, for the particular contract or order. All non-conforming cable, or cord which contains a MIL-C-915 marker tape and which is determined to be unacceptable to the Government shall be recorded with NAVSEC. Such record shall be furnished to NAVSEC by the manufacturer within ten days of receiving notification of non-acceptance. The record shall include the cable or cord type and size, the basis for rejection, footage of each length and serial numbers of the marker tape from each end of each length.

4.5.3 Group A examination and tests. Group A examination and tests as required by the specification sheet shall be performed on samples selected in accordance with 4.5.2.1.

4.5.4 Group B tests. Group B tests as required by the specification sheet shall be performed on specimens taken from samples selected in accordance with 4.5.2.2.

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4.5.5 Group C tests. Group C tests as required by the specification sheet shall be performed on specimens taken from samples selected in accordance with 4.5.2.3. The number of conductors chosen from each sample and the number of specimens of insulation and jacket shall be the same as for group B tests (see 4.5.4).

4.5.6 Certified test reports. For all cables and cords requiring qualification the manufacturer shall furnish certified test reports to NAVSEC and to all original purchasers of these cables and cords. A copy of the test report shall be forwarded to NAVSEC upon shipment of the cable or cord from the factory. The test report shall contain the following information:

(a) Manufacturer's QPL number and date or serial number and date of NAVSEC letter of approval.
(b) A statement to the effect that the product was constructed from materials listed on the manufacturer's approved details of construction sheet.
(c) A statement that the product meets all of the requirements of this specification.
(d) Results of all specification conformance tests showing actual values obtained.
(e) Year and month cable or cord was manufactured.
(f) Serial numbers of the marker tape taken from each end of each length of cable or cord.
(g) Customer's name and contract or order number.

4.6 Comparison inspection. Comparison inspection is testing performed at the Government's expense and at a NAVSEC designated laboratory. The purpose of comparison inspection is to evaluate and compare constructions and test results to specification and qualification requirements.

4.6.1 Sampling procedure for comparison inspection. Once a manufacturer's product has been qualified, the manufacturer shall submit one specimen for comparison inspection from the first lot of each type and size that he produces. Thenceforth, the manufacturer shall submit one specimen from every fourth lot of each type and size that he produces unless there is a period of one year or longer between any two successive lots of the same type and size in which case, a specimen from the next lot of the same type and size shall be submitted. The entire lot of cable or cord represented by the specimen submitted for comparison inspection shall be retained by the manufacturer until notice of inspection results are received from the Government.

4.6.1.1 Length of specimens. The length of specimens required for comparison inspection shall be as follows:

Length	Cable Types
15 feet for	2AU, 3U, 2SU, 1SU, 3SU, 2U, MU, MS, 2SJ, 3SJ, 4SJ and 1SAU. All sizes.
15 feet for	SRW, DRW, TRW, PBTMU, ECM, TSP, MDU, TTOP, MMOP, DCOP, TCOP, DLT. All sizes.
20 feet for	DSWS, MCSF, MHOF, DHOF, THOF, FHOF, CVSF, JAS, TPU, 3SWU, 2SWU, 1SWU, 2WAU, 3SF, 1SMWU, 2SWAU, DNW, TNW, FNW, MNW, TPNW, 1SMU. All sizes.
25 feet for	All silicone rubber insulated cables having conductor size 30 and smaller; 1S50MU, 1S75MU, MCOS, MSP, MSPW.
35 feet for	1SWF, 2SWF, MWF, S2S. All sizes.
38 feet for	All silicone rubber insulated cables having conductor size 40 and larger.
# 50 feet for	DSS, TSS, FSS, 5SS, 7SS, TTRS. All sizes.

4.6.1.2 Marking of specimens. All test specimens shall be marked with the following information:

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4.6.1.3 Packaging of specimens for shipment. To facilitate handling, specimens may be shipped in boxes or wrapped coils.

4.6.1.4 Notice of shipment. When specimens are shipped to a laboratory for comparison inspection, notice of such shipment shall be sent separately to the laboratory with a copy to NAVSEC. The notice shall contain the same information as required for marking of the specimen.

4.6.2 Action in case of failure. If any specimen fails one or more of the specified comparison inspection tests, the lot represented by that specimen shall be held in abeyance. The manufacturer shall have the option of resubmitting the lot, but only after the lot has been re-evaluated, the cause of failure determined, and corrective action has been taken. If the manufacturer resubmits the lot, the specimen representing the lot shall be subjected to the test or tests under which failure occurred plus any other tests which NAVSEC may deem necessary to assure compliance with this specification. If the manufacturer chooses not to resubmit the lot, or if the lot is resubmitted and a failure is found, the lot shall be rejected. Once comparison inspection reveals a failure in a particular type and size subsequently produced shall be submitted until two successive lots pass comparison inspection. The manufacturer shall then resume the normal rate of submission of samples for comparison inspection.

4.6.3 Action in case of rejection. Any cable or cord which requires qualification and which contains a MIL-C-915 marker tape and which has been rejected on the basis of comparison inspection and has been determined to be unacceptable to the Government shall be recorded with NAVSEC. Such record shall be furnished to NAVSEC by the manufacturer within ten days of receiving notification of non-acceptance. The record shall include the cable, or cord type and size, the basis for rejection, footage of each length and serial numbers of the marker tape from each end of each length.

4.6.4 Test reports. One copy of laboratory reports covering comparison inspection shall be forwarded to the cable or cord manufacturer and one copy shall be forwarded to NAVSEC.

4.7 Conductor strand inspection. Before individual strands are assembled into a conductor they shall be tested for coating when applicable, tensile strength and elongation. Test procedures shall be in accordance with the applicable ASTM standard referenced in 3.3.1. Lot sizes and number of specimens per lot shall be as follows:

Diameter of individual strand (inch)	Lot size	Number of test specimens
0.003 to .013	2500 pounds or less	2
	Each additional 2500 pounds or fraction thereof	1
.014 to .040	5000 pounds or less	2
	Each additional 5000 pounds or fraction thereof	1
.041 to .129	10,000 pounds or less	2
	Each additional 10,000 pounds or fraction thereof	1

Records of tests shall be kept, and copies of the applicable records shall become a part of the test reports for finished cable or cord.

4.8 Test methods (physical).

4.8.1 Abrasion resistance.

4.8.1.1 Specimens. Specimens shall be approximately 24-inch lengths of completed conductor (insulation plus coverings, if any). For cables or cords containing four conductors or less, one specimen shall be taken from each conductor. For cables or cords

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containing more than four conductors, the number of specimens shall be the square root (to the nearest whole number) of the total number of conductors, but in no case less than four.

4.8.1.2 Apparatus. The abrading machine shall consist of an 8-inch diameter metal drum with two abrading tools mounted 180 degrees apart on the circumference. Each abrading tool shall be a 5/16-inch square high-speed steel tool bit (Cleveland Twist Drill Co. #855 or equal). The tool bits shall be ground off so as to produce 90-degree sharp edges free of visible nicks, with not more than 0.004 inch per surface being removed. The grinding wheel used shall be medium grade (Macklin Co. #48A-H8-V6, or equal). The tools shall be secured in notches in the rim of the drum so that one diagonal through the cross-section of the bit is coincident with the diameter of the drum, and the other diagonal is tangent to the surface of the drum. The distance from the center of the drum to the longitudinal axis of the tool bit shall be 4 inches.

4.8.1.3 Procedure. Prior to testing, specimens shall be wiped with a clean, dry cloth to remove any lubricant or dirt. The test specimen shall be laid over the drum so as to extend approximately half way around the drum. One end of the specimen shall be secured to the base of the drum support, and a 1-pound weight fastened to the other end. The direction of rotation of the drum shall be away from the fixed end of the specimen, toward the weighted end. Rate of rotation shall be approximately 17 revolutions per minute. A voltage from 12 to 120 volts shall be applied between the abrading tools and the conductor of the specimen.

4.8.1.4 Observation. Electrical contact between the abrading tools and the conductor, before the required number of scrapes have been performed, shall constitute failure of the specimen.

4.8.2 Accelerated service.

4.8.2.1 Specimens. A specimen shall consist of a length of completed cable 15 feet long.

4.8.2.2 Apparatus. The test voltage (see 4.8.2.3) shall be 60 Hz, and shall be supplied by either one three-phase transformer or three single-phase transformers. Transformer minimum voltage rating shall be equal to the test voltage; minimum current rating shall be 3 amperes when testing cable rated at 1 kv, and 1/2 ampere when testing cable rated at 5 kv. The loading current shall be supplied by three single-phase loading transformers having the secondary winding isolated from ground. Instruments for measuring voltage and current to an accuracy of plus or minus 2 percent shall be provided.

4.8.2.3 Procedure. The specimen shall be formed into a large loop within the test chamber, in a horizontal position so as to permit free circulation of convection air currents. There shall be means of preventing drafts of air from directly striking the specimen. Each phase of the test voltage shall be connected to the conductors of the specimen through a fuse. The fuses shall be normal-blow 1-ampere for cables rated for 1 kv, and normal-blow 1/4-ampere for cables rated for 5 kv. (In lieu of fuses, overload relays having equivalent tripping characteristics may be used). The specimen shall be heated by the circulation of the loading current specified on the specification sheet. Air temperature within the test chamber shall be maintained at 30° +2°C. during the test. The test voltages and their points of application shall be as follows:

(a) Cables rated for 5 kv:
  (1) 4160 volts from conductor to conductor
  (2) 2400 volts from conductor to armor (when specimen is armored)
(b) Cables rated for 1 kv:
  (1) 450 volts from conductor to conductor
  (2) 260 volts from conductor to armor (when specimen is armored)

(Four-conductor cables shall have the green conductor either isolated or grounded to the armor when the specimen is armored, and shall be treated as three-conductor cables. For six-conductor cables, the physically opposite conductors shall be connected in parallel to form three pairs, and shall be treated as three-conductor cables. For all sizes of cable, the voltage and the loading currents shall be maintained continuously for a period of 7 hours, unless dielectric failure occurs before the expiration of this period. Measurement of the maximum and average temperatures of the cable jacket shall be accomplished by means of four thermocouples placed on the top of the specimen, at 2-inch intervals near the mid-point of the specimen.)

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4.8.2.4 Observation. Blowing of any fuse or tripping of any overload relay, before the expiration of the 7-hour period, shall constitute a failure of the specimen. Also, a jacket temperature exceeding 135°C., exudation of the jacket through the armor (when specimen is armored), or the sagging away of the cable jacket from the foundation of the specimen, within the 7-hour test period, shall constitute failure of the specimen. (Exudation is defined as actual flow or expansion of the jacket to such an extent that the armor cuts through the jacket material. Bulging of the jacket through the armor, without actual flow or cutting, is not considered exudation.)

4.8.3 Armor.

4.8.3.1 Physical. Specimens of the armoring wire shall be subjected to tests of tensile strength, and elongation, springiness, and toughness before application to the cable.

4.8.3.1.1 Springiness. A length of armoring wire shall be coiled snugly around a 1/4-inch mandrel for several turns with a tension of not less than 100 grams nor more than 125 grams, and then released. The outside diameter of the coil, as measured in several places at or near its middle, shall not exceed 3/8-inch.

4.8.3.1.2 Toughness. The toughness of the wire shall be determined by bending a specimen back and forth, in a specially designed fixture, over a 0.030-inch radius support. Each 90-degree movement in either direction shall be taken as one bend. The bends shall be made under a uniform tension of 100 grams, and at a uniform rate not to exceed 50 bends per minute. The wire shall withstand a minimum of ten 90-degree bends before breaking.

4.8.4 Bending endurance.

4.8.4.1 Specimen. A specimen shall consist of a 30-inch length of completed cable or cord.

4.8.4.2 Procedure. All conductors of the specimen shall be connected so as to form a single series circuit for the purpose of detecting conductor breakage during the test. The specimen shall be suspended vertically between two sets of rollers with the top end secured to a power driven reciprocating arm. Except at the arm clamp and between each set of rollers, the specimen shall be not closer than 1/4 inch to any machinery part. The specimen shall be kept under a tension of 40 ±2 psi of the specimen's cross-sectional area by attaching a weight to the lower end. The cross-sectional area of the specimen shall be determined by the maximum diameter for the cable or cord type and size as given by the specification sheet. The tension force (clamping device plus weight) shall be directly downward. The clamping devices used to secure the weight to the specimen and the specimen to the arm shall be such as to apply uniform radial pressure to the specimen to prevent slippage of the specimen or any of its parts during the test. The distance from the axis of rotation of the arm and the nearest surface of the arm clamp shall be 9 ±1/4 inches. The top set of cylindrical metal rollers over which the specimen will be bent shall have smooth sides, shall have the diameter as required by the specification sheet and shall be mounted so as to be free-rotating at all times. The axis through the centers of these rollers shall be horizontal and the horizontal distance between roller faces shall be equal to the maximum overall diameter of the specimen to be tested plus 1/8 inch. The rollers shall be equidistant from the vertical centerline through the axis of rotation of the reciprocating arm. The vertical positioning of the top pair of rollers shall be such that the distance between a horizontal centerline through the axis of rotation of the arm and the top of the rollers is equal to one-half the maximum diameter of the specimen. The vertical positioning of the lower or guide rollers shall be approximately midway between the top rollers and the bottom clamp. The bottom rollers shall be not less than 3/8 inch in diameter and shall have the same horizontal spacing as the top rollers. The entire assembly (equipment plus specimen) shall be conditioned in a chamber to the temperature specified on the specification sheet ±2°C. While at this temperature, the specimen shall be bent, at approximately its mid-point, over the rollers at the rate of 12 to 14 cycles per minute. One cycle shall consist of the reciprocating arm swinging through an arc of 180 degrees and returning (360 degrees total swing). After completion of the number of cycles and at the temperature specified by the specification sheet, the specimen shall be removed from the test apparatus and subjected to the voltage withstand test specified by the specification sheet.

4.8.4.3 Observation. After completion of the voltage withstand test, the specimen shall be visually examined, inside and out. Any electrical breakdown, breaking of conductor strands or cracking of insulation or jacket material shall constitute failure. If failure occurs within 2 inches of the grip on either end of the specimen, the results shall be disregarded and the test repeated.

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4.8.5 Breaking strength.

4.8.5.1 Specimen. A specimen shall consist of a length of completed cable or cord approximately 6 feet long.

4.8.5.2 Apparatus. The testing machine shall be power-driven, and shall be equipped with a dial, scale, or automatic recorder that will record within plus or minus 1 percent of the force required to break the specimen.

4.8.5.3 Procedure. One end of the specimen shall be wrapped two full turns around the upper mandrel and secured in place to provide sufficient snubbing action to prevent slipping. The other end of the specimen shall be secured in a similar manner to the lower mandrel. The distance between the jaws with the specimen in place shall be not less than 6 inches. The jaws shall be separated at a uniform rate of approximately 1 inch per minute until the specimen breaks.

4.8.5.4 Observation. The breaking strength of the specimen shall be not less than that required by the specification sheet.

4.8.6 Cable aging (260°C.).

4.8.6.1 Specimen. A specimen shall consist of a 4-foot length of completed cable.

4.8.6.2 Procedure. The specimen shall be coiled or bent into a "U" shape and secured. The inside radius of the coil or "U", measured to the nearest 1/4-inch, shall be six times the diameter of the specimen. The specimen shall be suspended vertically, with the ends extending downward, in an air oven which has forced circulation of fresh air. The oven temperature shall be raised gradually, in a 2 to 3-hour period, from room temperature to 260°C. and held at 260° ±10°C. for a period of 24 hours and then returned to room temperature. When the oven has reached room temperature, the specimen shall be removed and examined visually. After visual examination of the cable specimen is made, tensile strength and elongation measurements shall be made on specimens of the jacket removed from the aged cable specimen.

4.8.6.3 Observation. The falling away of any material in excess of 1 gram from the ends of the cable specimen, during the time the specimen is in the oven shall be cause for rejection. Any flow or expansion of the jacket through the armor to such an extent that the armor cuts into the jacket material shall be cause for rejection. (Expansion of the jacket without actual cutting of the armor into the jacket shall not be considered cause for rejection). Shrinkback of the jacket in excess of 1/8-inch from either end of the specimen shall be cause for rejection. Jacket specimens cut from the aged cable specimen shall exhibit a tensile strength of not less than 600 psi and an elongation of not less than 150 percent.

4.8.7 Cable aging and compatibility (95°C.). This test is applicable to cables having thermoplastic insulation and is for the purpose of detecting any significant degradation in the characteristics of the cable or its parts, resulting from either part incompatibility or prolonged overheating.

4.8.7.1 Specimens. Two specimens of completed cable are required. Specimen number 1 shall be approximately 30 feet long. Specimen number 2 shall have an approximate length of 2 feet plus 24 times the overall diameter of the cable. Both specimens shall be taken from one sample length of cable.

4.8.7.2 Apparatus. The test apparatus shall consist of a heat chamber, a means of measuring and controlling the chamber temperature, a recording thermometer for chamber temperature, and racks or other means of supporting the specimens within the chamber. The temperature control shall be capable of maintaining the chamber temperature at 95° ±3°C. for a continuous period of 400 hours. Additional apparatus shall consist of a mandrel having a diameter of approximately 12 times the specimen diameter.

4.8.7.3 Procedure. The procedure shall consist of two parts: (1) heat aging; (2) bending.

4.8.7.3.1 Part one - heat aging. Prior to aging, Specimen number 1 shall be given the voltage withstand and the insulation resistance tests. The maximum insulation resistance values measured for each conductor shall be recorded. The two specimens shall be placed into the chamber in a horizontal position. Specimen number 2 shall be kept straight during the aging. The two specimens shall be conditioned at 95° ±3°C. continuously for

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400 hours. At the end of this period they shall be allowed to cool to room temperature. Specimen number 1 shall again be given the voltage test and the insulation resistance test. The maximum insulation resistance values obtained for each conductor shall again be recorded. Then two additional specimens shall be cut from Specimen number 1. One of these shall be given the watertightness test; the other shall be given the cable filler removability test.

4.8.7.3.2 Part two - bending. With one end of Specimen number 2 secured to the mandrel, or directly adjacent thereto, force shall be applied to the free end so as to bend the specimen over the mandrel through an arc of approximately 180 degrees, at a rate of approximately 20 degrees per second. The specimen shall be secured so as to maintain it in this bent position, after which it shall be dissected and given a visual examination of all components.

4.8.7.4 Observations. Any of the following results shall be cause for rejection.

(a) Failure of the specimen to meet the requirements of the cable filler removability test.
(b) Failure of Specimen number 1, after heat aging, to meet the value of voltage withstand required by the applicable specification sheet.
(c) Insulation resistance values for Specimen number 1, after heat aging, which are below the minimum values specified on the applicable specification sheet
(d) Any dripping of the filler material away from the ends of the specimens during the heat aging.
(e) Any change in either the filler material or any part with which it is in contact, such as hardening, discoloration or other visible deterioration, of a nature or to an extent as to impair the performance of the cable in service.

4.8.8 Cable aging and compatibility (125°C.). This test is applicable to cables having either thermosetting insulation or silicone rubber-glass tape insulation, and is for the purpose of detecting any significant degradation in the characteristics of the cable or its component resulting from either component incompatibility or prolonged overheating.

4.8.8.1 Specimens. A specimen shall consist of a length of completed cable approximately 40 feet long.

4.8.8.2 Apparatus. The test apparatus shall consist of a heat chamber, a means of measuring and controlling the chamber temperature, a recording thermometer for chamber temperature, an internal mercury thermometer, and a rack or other means of supporting the specimen within the chamber. The temperature control shall be capable of maintaining the chamber temperature at 50° +3°C. for a continuous period of 400 hours. There shall also be a source of current capable of supplying the required amount of current for a continuous period of 400 hours. There shall also be a mandrel having a diameter of approximately 12 times the specimen diameter.

4.8.8.3 Procedure. The procedure shall consist of two parts: (1) heat aging; (2) bending.

4.8.8.3.1 Part one - heat aging. Prior to heat aging, the specimen shall be given the voltage withstand and the insulation resistance tests. The maximum values of insulation resistance measured for each conductor shall be recorded. During aging, conductor temperature shall be determined either by direct measurement or by the measurement of change of copper resistance. For the direct measurement method, a thermocouple shall be inserted through a small knife puncture in the cable, approximately midway between the ends of the specimen. For cables having only one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor. For cables having more than one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor in the innermost layer. Thermocouples shall also be attached to the jacket of unarmored cables. In the case of armored cables, the thermocouples shall be attached to the armor. The specimen shall be placed into the chamber so as to have a straight section of approximately 7 feet, and so as to permit free circulation of convection air currents. The amplitudes of loading currents shall be such as to maintain each conductor at a temperature of 125° ±5°C. continuously for 400 hours. Readings of time, chamber temperature, jacket or armor temperature, and the current in each conductor shall be recorded at 15-minute intervals during the first hour, at 1-hour intervals during the next 5 hours, and twice daily thereafter for the duration of the test. Record shall also be made of any gas or fumes emanating from the specimen, and of any oozing of material from the ends of the specimen. In the case of armored cable, record shall also be made of any exudation of jacket material through the armor. At the end of the 400-hour test period, the

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specimen shall be allowed to cool to room temperature. Then it shall be given the voltage withstand and the insulation resistance tests. The maximum insulation resistance values measured for each conductor shall again be recorded. From the portion of the 40-foot specimen which was kept straight during the heat aging, specimen having an approximate length of 2 feet plus 24 times the overall diameter of the specimen shall be taken. This specimen shall be used for the bending procedure. From the remainder of the 40-foot specimen, two additional specimens shall be taken. One of these shall be given the watertightness test; the other shall be given the cable filler removability test.

4.8.8.3.2 Part two - bending. With one end of the specimen secured to the mandrel, or directly adjacent thereto, force shall be applied to the free end so as to bend the specimen over the mandrel through an arc of approximately 180 degrees, at a rate of approximately 20 degrees per second. The specimen shall be secured so as to maintain it in this bent position, after which it shall be dissected and given a visual examination of all parts.

4.8.8.4 Observations. Any of the following results shall be cause for rejection.

(a) Failure of the specimen to meet the requirements of the cable filler removability test.
(b) Failure of the 40-foot specimen, after heat aging, to meet the value of voltage withstand required by the applicable specification sheet.
(c) Insulation resistance values for the 40-foot specimen, after heat aging, which are below the minimum values specified on the applicable specification sheet.
(d) Any dripping of the filler material away from the ends of the 40-foot specimen, during the heat aging.
(e) Any change in either the filler material or any part with which it is in contact, such as hardening, discoloration or other visible deterioration, of such a nature or to an extent as to impair performance of the cable in service.

4.8.9 Cable filler removability. This test is applicable to the non-fibrous filler material used to fill the voids between insulated conductors or groups of insulated conductors. It is not applicable to filler material used to fill the voids between the strands of conductors or the strands of shields.

4.8.9.1 Specimens. A specimen shall consist of a length of completed cable or cord approximately 2 feet long.

4.8.9.2 Procedure. All overall cable or cord parts, such as armor, jacket, binders, shield, etcetera shall be removed from one end of the specimen so as to expose the cable or cord core and the filler material, for a distance of about 1 foot. Using fingers only (no hand tool), the filler shall be separated from the conductors or groups of conductors for approximately their fully exposed length. In the case of a group of conductors such as a pair or triad which has a common covering such as a shield or a binder tape, this covering shall be removed so as to expose the conductors and the filler material, for a distance of about 6 inches. The filler material shall be separated from the conductors as was done in the preceding step. For cables or cords having more than one such group, the test shall be performed on not less than one group.

4.8.9.3 Observation. The filler material shall be flexible and shall be removable from any part with which it is in contact, through the use of fingers only. It shall not adhere to the fingers, to the insulation of conductors, or to tapes or other coverings over shields. The presence of occasional particles or slivers of filler residue will be acceptable, providing that these can be removed by light brushing with the fingers or with a dry cloth. Filler material which leaves residue that is removable only by vigorous wiping or through the use of solvents will not be acceptable.

4.8.10 Cold bending, cable.

4.8.10.1 Specimen. A specimen shall consist of a 5-foot length of completed cable.

4.8.10.2 Procedure. The specimen shall be straightened and, if necessary, kept straight by securing to a straight wooden bar. The straight specimen shall be placed in a cold chamber and conditioned for a minimum of 6 hours. The conditioning temperature shall be minus 20° +2°C., unless otherwise specified by the specification sheet. The specimen shall then be removed from the conditioning chamber and, without delay, bent 180 degrees around a wooden mandrel of the specified diameter (see specification sheet) at a rate of from 10 to 20 degrees per second. The specimen shall be secured so as to maintain this bent position.

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4.8.10.3 Observation. While the specimen is in the bent position, it shall be dissected and a visual examination made of all its parts. When defects or disarrangements are discovered in any part, examination shall be made of specimens in the "as received" condition to determine whether the defect existed prior to the cold bend test. Any defects, such as torn tapes or broken or cracked parts of a nature or to an extent as to impair the performance of the cable, in service, shall be cause for rejection.

4.8.11 Cold bending, cord.

4.8.11.1 Specimens. Two specimens of completed cord, each approximately 20 inches long, are required.

4.8.11.2 Apparatus. The apparatus shall consist of a refrigerator chamber, temperature controls, mandrels and magnifying glass. The chamber shall be large enough to permit vertical suspension of the specimen. The chamber shall be equipped with a device to hold a mandrel, with specimen attached, in such a manner that the specimen can be wrapped around the mandrel without touching either the mandrel or the specimen with the hands. The temperature controls shall be capable of maintaining a chamber temperature of minus 40° +2°C. for a period of 20 hours. The mandrels for testing the complete cord shall have a diameter not greater than three times the specified minimum diameter of the specimen. The mandrels for testing the insulated conductors shall have a diameter not greater than the nominal diameter over the insulated conductor. The magnifying glass shall provide at least 3-diameter magnification.

4.8.11.3 Procedure. The specimen selected for testing the complete cord shall be attached at one end to the proper size mandrel. The specimen selected for testing the insulated conductors shall have the conductors removed therefrom and each insulated conductor shall be attached at one end to the proper size mandrel. All specimens shall be suspended vertically within the chamber with their lower ends weighted sufficiently to keep the specimens taut and to permit wrapping them without handling. The temperature within the chamber shall be reduced to minus 40° +2°C and held at this temperature for a period of 20 hours. At the end of 20 hours and while the specimens are still in the chamber at minus 40° +2°C., they shall be wrapped for five close turns around their respective mandrels and then secured to the mandrels in the wrapped position. The specimens and mandrels shall then be removed from the chamber and the specimens examined through a magnifying glass of at least 3-diameter magnification.

4.8.11.4 Observation. Evidence of cracks, flaws or other damage caused by the cold bend procedure shall constitute failure.

4.8.12 Cold working (minus 20°C.).

4.8.12.1 Specimen. A specimen shall consist of a 20-foot length of completed cable.

4.8.12.2 Procedure. The specimen shall be conditioned in a chamber at minus 20° +2°C. for a period of not less than 6 hours. After the conditioning period the specimen shall be removed from the chamber and immediately shaped around two rollers, each 12 to 13 times the diameter of the cable. The specimen shall be shaped and pulled over the pulleys in such a manner that an "S" bend of 180 degrees is imposed progressively on the entire length of the specimen. The pull shall be done once in each direction at the rate of 35 ±5 feet per minute.

4.8.12.3 Observation. After the cold working, the specimen shall be dissected and visual examination shall be made of all parts. When any defects or disarrangements are discovered, examination shall be made of a specimen in the "as received" condition to determine whether the defect or disarrangement existed prior to the cold working. Any defects such as torn tapes or broken or cracked parts of a nature or to an extent as to impair the performance of the cable in service shall be cause for rejection.

4.8.13 Cold working (minus 54°C.).

4.8.13.1 Specimen. A specimen shall consist of a 20-foot length of completed cable or cord.

4.8.13.2 Procedure. A specimen shall be conditioned in a chamber at minus 54° +2°C. for a period of not less than 6 hours. After the conditioning period and while still in the chamber at minus 54° +2°C., the specimen shall be shaped and pulled around two 12-inch diameter grooved sheaves in such a manner that an "S" bend of 180 degrees in each direction is imposed progressively on the entire length for a total of four times in each direction. The specimen shall be pulled over the sheaves at the rate of 50 ±5 feet per minute.

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4.8.13.3 Special procedure (for qualification). When the cold working (minus 54°C.) test is being performed for qualification purposes additional steps shall be taken as follows:

(a) Immediately prior to subjecting the specimen to minus 54° ± 2°C. for 6 hours, precondition the specimen by soaking in salt water (approximately 3 percent salt by weight) at a temperature of 22° ±5°C. for a period of not less than 21 days.
(b) The number of cycles for pulling the specimen over the sheaves shall be ten.
(c) After the ten cold-working cycles and after the specimen has returned to room temperature, the center 5-foot section shall be removed and subjected to the hydrostatic (open end) test.

4.8.13.4 Observation. After the cold working, the specimen shall be removed from the chamber and dissected. Visual examination shall be made of all parts. When any defects or disarrangements are discovered, examination shall be made of a specimen in the "as received" condition to determine whether the defect or disarrangement existed prior to the cold working. Any defects such as torn tapes or broken or cracked parts of a nature or to an extent as to impair the performance of the cable or cord in service shall be cause for rejection. In addition, when the test has been performed for qualification, water leakage in excess of that permitted by the specification sheet or slippage of the core in excess of 1/4-inch shall also be cause for rejection.

4.8.14 Crack resistance.

4.8.14.1 Specimens. Specimens shall consist of lengths of completed conductors (insulation plus coverings, if any), which have been taken from completed cable. For cables containing four conductors or less, one specimen shall be taken from each conductor. For cables containing five or more conductors, the number of specimens taken (one per conductor selected), shall be the square root (to the nearest whole number) of the total number of conductors in the cable; in no case shall the number of specimens be less than four. The length of specimens shall be in accordance with 4.8.14.2.

4.8.14.2 Procedure. The test shall be performed at room temperature (23° ±3°C.). The specimen shall be hand-wound around a mandrel for three complete turns, unwound, wound around the mandrel for three complete turns in the opposite direction, unwound and straightened. The specimen shall then be immersed for 1/2 hour in an aqueous solution of gentian violet which is at room temperature. Mandrel diameters and specimen lengths shall be in accordance with the following or as specified in the individual specification sheet:

Conductor size (Navy Standard)	Mandrel diameter (inches, nom.)	Specimen length, minimum (inches)
23	1-1/2	24
14	1	18
9	1/2	12
4	1/2	12
3	1/3	8
2	1/3	8
Smaller than 2	1/4	6

4.8.14.3 Observation. Any visible damage to the braid, or any cracking or peeling of the braid or insulation covering, such as to permit the penetration of fluid to the dielectric, shall be cause for rejection. The number of pinholes in the braid covering shall not exceed an average of one per linear foot of conductor. When necessary to more accurately determine this limit, a 10-foot long specimen shall be subjected to immersion after bending.

4.8.15 Drip.

4.8.15.1 Specimen. A specimen shall consist of an 18-inch length of completed cable.

4.8.15.2 Procedure. The specimen shall be suspended vertically in an oven for a period of not less than 18 hours. Over temperature shall be maintained within plus or minus 1°C. of the temperature specified on the specification sheet.

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4.8.15.3 Observation. The falling away of any material from the lower end of the specimen, during the time the specimen is in the oven, shall be cause for rejection.

4.8.16 Flammability.

4.8.16.1 Specimens. Specimens shall consist of completed cable or cord approximately 18 inches in length.

4.8.16.2 Apparatus. The apparatus shall consist of an enclosure, supports, heater coil, spark plugs, flame travel gage, and associated accessories.

4.8.16.2.1 Enclosure. An enclosure of sufficient size to contain the specimen, supports, heater coil, spark plugs, flame travel gage, and their associated accessories shall be arranged to eliminate air drafts and permit clear view of the interior through shatterproof glass windows. Vent holes, distributed around the sides adjacent to the base, shall be provided to admit fresh air when an exhaust fan connected to the top of the enclosure is operated at a minimum suction just sufficient to carry off smoke and gases.

4.8.16.2.2 Supports. Supports shall be suitable for holding the specimens in a vertical position with an unsupported span not less than 14 inches long. At the lower end of the specimen, means shall be provided for containing the gases expelled from the specimen so that they do not diffuse throughout the test enclosure, but instead are deflected upward toward the spark plugs. (A simple cup-shaped device, placed just under the specimen, should be suitable for this purpose).

4.8.16.2.3 Heater coil. Heater coils shall consist of seven turns of 0.102-inch diameter resistance wire conforming to composition E of QQ-R-175, space wound to 0.25-inch between centers. The inside diameter of the coil shall exceed the overall diameter of the specimen by not less than 0.5 inch nor more than 0.6 inch. The lower end of the heater coil shall be located 1.5 inches above the top of the lower specimen support.

4.8.16.2.4 Spark plugs. Two spark plugs, with extended electrodes spaced 1/8-inch from the surface of the specimen and the plugs located on diametrically opposite sides of the specimen, shall be placed with their longitudinal certerlines in a horizontal plane 1/2-inch above the top of the heater coil, to ignite the gases emitted from the heated specimen. An electric circuit adequate to maintain continuous sparking at the electrodes during the specified time shall be provided. The spark plugs shall be mounted in such a manner that they may be moved away from the specimen after ignition takes place so as not to impede the travel of the flame and to prevent their electrodes from becoming fouled by soot.

4.8.16.2.5 Flame travel gage. A flame travel gage shall be positioned near the specimen to judge the distance of flame travel without appreciably impeding the progress of the flame.

4.8.16.3 Procedure. With the enclosure closed and ventilated, the specimen centered in the heater coil, and the spark plugs and flame gage properly located, a stop watch shall be started simultaneously with the energizing of the heater coil and spark plugs. A constant current as specified in table XII shall be supplied from a transformer source to the heater coil. Ignition shall be considered as occurring when the flame transfers from the escaping gases to the surface of the specimen and continues there, disregarding the flashes which may occur in the gassing space prior to the sustained flame. The time required to ignite the specimen shall be recorded. For all armored cable, the heating current shall be turned off 30 seconds after ignition occurs. When armored cable does not ignite in 250 seconds or more, the specimen is considered satisfactory, and the test shall be discontinued. For all unarmored cables or cords where ignition time is less than 60 seconds, the heating current shall be turned off 60 seconds after being turned on; where the ignition time is greater than 60 seconds, the heating current shall be turned off when ignition occurs. Immediately after ignition occurs, the electrical supply to the spark plugs shall be cut off, and the plugs shifted away from the flame. The maximum distance to which the flame travels along the surface of the specimen, measured from the top of the heater coil, before extinction shall be noted and recorded.

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Table XII - Current in heating coil.

Nominal diameter of cable or cord (inches)	Current amperes a.c. (rms)	Nominal diameter of cables or cord (inches)	Current amperes a.c. (rms)
0.050 thru 0.099	45	0.80 thru 0.89	53
.100 thru .19	46	.90 thru .99	54
.20 thru .29	47	1.00 thru 1.09	55
.30 thru .39	48	1.10 thru 1.39	56
.40 thru .49	49	1.40 thru 1.79	57
.50 thru .59	50	1.80 thru 2.29	58
.60 thru .69	51	2.30 to 3.00	59
.70 thru .79	52		

4.8.16.4 Observation. The number of seconds that the specimen continues to burn until the cessation of all flaming, after the current in the heater coil has been cut off, shall be recorded as the time required for self-extinction. If a specimen tested fails to comply with any of the limitations specified in table XIII or burns for a distance greater than that permitted by the specification sheet, upon the first trial, two additional specimens shall be tested, and the results of all three trials averaged together. Specimens not passing the flammability limits shall constitute a failure.

Table XIII - Flammability limits

Cable or cord construction or type	Ignition time, minimum (seconds)	Burning time, maximum (seconds)
With armor	60	60
Without armor (except MRI)	30	120
MRI	25	60

4.8.17 Gas flame.

4.8.17.1 Specimen. A specimen shall consist of a length of completed cable approximately 4 feet long, with the ends prepared for electrical connection.

4.8.17.2 Apparatus. The test chamber shall be equipped with means for exhausting gases resulting from burning, without causing horizontal air currents which prevent a steady, even gas flame. The power supply shall be of the three-phase, four-wire type, capable of supplying 3 amperes minimum. (Instruments for measuring current, capacitance and power factor are required at the Government laboratory only). The source of flame shall be a 24-inch ribbon type gas burner designed to produce a flame which is uniform in height throughout the length of the burner. Means of metering both air and gas inputs to the burner shall be provided.

4.8.17.3 Procedure. The specimen shall be suspended horizontally over the burner and parallel to it, so that the underside of the specimen is 1-3/4 to 2 inches above the top surface of the burner. The specimen shall be supported adequately so that during the time it is exposed to flame, no part of it may sag closer than 3/4 inch from the top of the burner. The chamber shall permit free circulation of convection air currents. The standard flame shall be that produced by combustion of 27.5 cubic feet per hour of natural gas having a heat content of 1030 ±2 percent British thermal units (Btu) per cubic foot, or another gas having the equivalent heat output and temperature. The conditions for measurement of gas and air flow rates shall be atmospheric pressure (14.7 psi) and 23°C. If measured at other pressures or temperatures, the values shall be corrected to the above specified conditions. **The transformer shall be connected to the conductors of the** specimen through a fuse in each phase leg. The fuses shall be normal-blow 1-ampere for cables rated for 1 kv and normal-blow 1/4-ampere for cables rated for 5 kv. (In lieu of fuses, overload relays having equivalent tripping characteristics may be used.) The gas flame shall be applied continuously for the duration of the time period specified by the specification sheet, unless failure of the specimen occurs first. The test voltages and their points of application shall be as follows:

(a) Cables rated for 5 kv:
(1) 4160 volts for conductor to conductor

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(2) 2400 volts from conductor to armor (when specimen is armored)
(b) Cables rated for 1 kv and having conductor sizes 14 and larger:
(1) 450 volts from conductor to conductor
(2) 260 volts from conductor to armor (when specimen is armored)
(c) Cables rated for 1 kv and having conductor sizes smaller than 14:
(1) 120 volts from conductor to conductor
(2) 70 volts from conductor to armor (when specimen is armored)

(For four-conductor and six-conductor cables, the physically opposite conductors shall be connected in parallel and the cables treated as two-conductor and three-conductor cables, respectively. Cables having seven or more conductors shall have adjacent conductors connected to different phases. In a 10-conductor cable, for example, conductors 2, 4, 6 and 9 should be connected to phase A; conductors 3, 5 and 8 should be connected to phase B; conductors 1, 7 and 10 should be connected to phase C. When the specimen is armored, the armor should be connected to the system neutral).

4.8.17.4 Observation. Blowing of any fuse or tripping of any overload relay, before the expiration of the period of time specified on the specification sheet, shall constitute failure of the specimen.

4.8.18 Heat distortion.

4.8.18.1 Specimens. Specimens for conductor sizes 200 MCM and smaller shall consist of 1-inch lengths of insulated conductor. Specimens for conductor sizes larger than 200 MCM shall consist of sections of insulation which have been prepared by removing the insulation from approximately 8-inch lengths of conductor, buffing strips of the insulation to a thickness of 0.050 ±0.010 inch and then cutting the buffed insulation into sections which are 1-inch long and 9/16 ±1/16-inches wide.

4.8.18.2 Apparatus. The apparatus shall consist of a small oven with controls capable of maintaining a constant temperature of 121° ±1°C. for a period of 3 hours, a Randall and Stickney gauge, or equivalent gauge having a 3/8-inch foot which weighs approximately 85 grams, and weights for loading the gauge foot. The weights shall be in addition to the weight of the gauge foot and shall be as follows:

Specimen or conductor size	Load for gauge (Grams)
3	400
4 thru 14	500
23 thru 75	750
100 thru 200	1000
Specimens from sizes larger than 200	2000

4.8.18.3 Procedure. The procedure for determining the percentage of distortion shall be as follows:

(a) Determine T_1, the thickness of the section of insulation to be tested.
Where the specimen consists of a buffed strip of insulation, use a Randall Stickney, or equivalent, gauge which has a 3/8-inch foot with no load weight of 85 grams. Where the specimen consists of a 1-inch length of insulated conductor, use a micrometer caliper which has flat surfaces for both the anvil and spindle, and measure the diameter over and under the insulation and designate these values D and C respectively.
Determine the thickness of the insulation, T_1, by the formula $T_1 = \frac{D-C}{2}$

(b) Preheat the oven to 121° ±1°C. and maintain it at this temperature for the remainder of the test.
(c) Place the gauge, which has been weighted in accordance with 4.8.18.2, into the preheated oven.
(d) At the end of 1 hour, place the specimen into the oven with the weighted gauge.
(e) At the end of the second hour, place the specimen under the weighted foot of the gauge.
(f) At the end of the third hour, read the dial of the gauge. Designate the value as T_2 when the specimen consists of a buffed strip of insulation.
Designate the value as F when the specimen consists of a 1-inch length of insulated conductor and calculate T_2 by the formula $T_2 = \frac{F-C}{2}$.
(g) Calculate the percent of distortion by the formula,

$$\text{Percent distortion} = \frac{T_1 - T_2}{2} \times 100.$$

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4.8.18.4 Observation. Distortion which exceeds 30 percent for specimens which consist of buffed strips of insulation or 20 percent for specimens which consist of a 1-inch length of insulated conductor shall constitute failure of the specimen.

4.8.19 Hydrostatic (open end).

4.8.19.1 Specimen. A specimen shall consist of a 5-foot length of completed cable or cord.

4.8.19.2 Procedure. One end of the specimen shall be fitted into the appropriate size of stuffing tube in accordance with MIL-S-24235/1. Water pressure shall be applied to the stuffing tube end of the specimen. The pressure and duration of application shall be as specified on the specification sheet covering the cable.

4.8.19.3 Observation. Unless otherwise specified by the specification sheet, no water leakage through the specimen will be permitted, and there shall be not more than 1/4 inch slippage of the cable or cord core or components.

4.8.20 Permanence of printing (conductor). The test shall be performed on a hori-
zontal abrasion machine, using a tape approximately 1/2-inch wide, comprised of No. 50-2/20 unbleached cotton braid. The tape shall be drawn back and forth across a portion of the specimen which bears the printed identification code legend. Motion of the tape shall be perpendicular to the axis of the specimen. The area of mutual contact between tape and specimen shall be such that the tape extends around the periphery of the specimen so that an arc of contact is not less than 135 degrees. A constant tension of 1/2 pound shall be maintained on one end of the tape. Total excursion of the tape across the specimen shall be 10 inches. A complete excursion of the tape across the specimen in one direction, followed by a complete excursion in the reverse direction to the starting point, shall constitute a cycle. The cyclic rate should be 28 $\pm$ 3 cycles per minute. No tape shall be used to perform more than 500 cycles per side.

4.8.20.1 Observation. Failure of the printed identification code legend to remain legible after completion of the number of cycles specified on the specification sheet shall constitute failure of the specimen.

4.8.21 Permanence of printing (jacket).

4.8.21.1 Specimen. A specimen shall consist of a length of completed cable approximately 6 inches long.

4.8.21.2 Apparatus. The apparatus shall consist of a motor-driven reciprocating mechanism capable of holding and moving a 0.025 $\pm$0.001 inch steel mandrel (usually a needle) in a horizontal motion back and forth for a distance of 3/8 inch on the surface of the specimen. In addition, there shall be a jig to hold the specimen at right angles to the mandrel and in a horizontal position and a counter to record the number of strokes.

4.8.21.3 Procedure. The specimen shall be mounted horizontally, at right angles to the mandrel with the printed legend on the top side. The mandrel shall be adjusted or weighted so as to exert a 1-pound force on the specimen throughout the test. The mandrel shall be rubbed along the printed legend for a distance of 3/8 inch and returned to original position to complete one cycle. The frequency of the cycles shall be 60 per minute. The specimen shall be subjected to a minimum of 250 cycles.

4.8.21.4 Observation. Failure of the printed legend to remain legible after completion of 250 rubbing cycles shall constitute failure of the specimen.

4.8.22 Physical tests on insulation and jacket. Tests shall be made on specimens of insulation and jacket to determine the physical properties of the materials before and after accelerated aging. Testing shall be performed on the aged and unaged specimens at the same time insofar as possible.

4.8.22.1 Specimens. Test specimens shall be approximately 6 inches long and shall be obtained from samples of completed cable. In the case of small conductors with thin walls, the specimen shall be the entire section of insulation. In the case of cable jackets or large conductors with thick walls, the specimen shall be either (a) a segment or section cut with a sharp knife held tangent to the conductor or cable core, or (b) a shaped specimen cut with a standard die. When the whole section is used the specimen shall not be cut longitudinally. The specimen shall be as free as possible from surface incisions and imperfections. In the event of surface irregularities, such as corrugations from stranding, the specimen shall be buffed to obtain smooth surfaces and uniform thicknesses. The cross-sectional area of the specimens shall be as large as possible but shall not exceed 0.030 square inch.

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4.8.22.1.1 Calculation of area. The area of specimens shall be calculated before aging. Calculation of the area of the test specimens shall be made as follows:

(a) Where the total cross section of the insulation is used, the area shall be taken as the difference between the area of the circle whose diameter is the minimum average outside diameter of the insulation and the area of the conductor. The area of a stranded conductor shall be figured from its maximum diameter. The following formula may be used to advantage:

$$A = 0.7854 (D + d) (D-d)$$

Where: A = Area of cross section in square inches.
D = Outside diameter of insulation in inches.
d = Diameter over conductor in inches.

(b) Where a slice cut from the insulation by a knife held tangentially to the wire is used, and the slice so cut has the cross section of a segment of a circle, the area shall be calculated as that of the segment of a circle whose diameter is that of the insulation. The height of the segment is the wall of insulation on the side from which the slice is taken. The values may be obtained from a table giving the areas of segments of a unit circle for the ratio of the height of the segment to the diameter of the circle.

(c) Where the cross section of the slice is not a segment of a circle, the area shall be calculated from the direct measurement of the volume or from the specific gravity and the weight of a known length of the specimen having a uniform cross section.

(d) Where a portion of sector or circle has to be taken as where the conductor is large and the insulation thin, the area shall be calculated as the thickness times the width. This applies either to a straight test piece or one stamped out with a die, and assumes that corrugations have been removed by buffing.

(e) Where a portion of a sector of a circle has to be taken, as where the conductor is large and the insulation thick, the area shall be calculated as the proportional part of the area of the total cross section.

4.8.22.1.2 Temperature. Specimens for both the aged and unaged physical tests shall be at a temperature of 25° ±3°C. and shall have been at this temperature for at least 30 minutes prior to testing.

4.8.22.1.3 Bench marks. All specimens shall be marked near the middle to provide a basis for determining elongation and set. The marks shall be 1 inch apart for polyproplyene and thermoplastic polyethylene insulations, and 2 inches apart for all other materials. Aged specimens shall be marked after aging.

4.8.22.2 Accelerated aging. Conditioning periods and temperatures for specimens to be aged shall be as shown in table XIV.

Table XIV - Accelerated aging of specimens.

Materials	Conditioning	Temperature (degrees C.)	Period (hours)
Insulations			
Polyethylene (thermoplastic)	Air oven	100	96
Polypropylene	Air oven	121	96
80°C. Polyvinyl chloride	Air oven	100	96
80°C. Semi-rigid Polyvinyl chloride	Air oven	113	168
105°C. Polyvinyl chloride	Air oven	136	96
Cross-linked polyethylene	Air oven	121	168
	Air pressure heat	127	40
Ethylene propylene rubber	Air oven	121	168
	Air pressure heat	127	40

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Table XIV - Accelerated aging of specimens (Cont'd).

Materials	Conditioning	Temperature (degrees C.)	Period (hours)
Insulations (Cont'd)			
Synthetic rubber	Air pressure heat	127	20
	Oxygen pressure	80	168
Butyl rubber	Air oven	100	168
	Air pressure heat	127	40
Jackets			
Special thermoplastic	Hot oil immersion	121	18
	Air pressure heat	126	20
Standard thermoplastic	Hot oil immersion	121	18
Polychloroprene	Hot oil immersion	121	18
	Air pressure heat	126	20
	Oxygen pressure	80	168
Chlorosulfonated polyethylene	Hot oil immersion	121	18
	Air pressure heat	126	20
	Oxygen pressure	80	168
Silicone rubber	Air oven	260	24

4.8.22.2.1 Air oven. Air oven aging shall be accomplished by heating the specimens at the required temperature and for the required period (see table XIV) in an oven which has forced circulation of fresh air. The oven temperature shall be maintained within plus or minus 2°C. of the specified temperature.

4.8.22.2.2 Air pressure heat. Air pressure heat aging shall be accomplished by suspending the specimens in a pressure chamber for the required period and at the required temperature (see table XIV). The required temperature shall be reached during the first 15 minutes of the period and maintained within plus or minus 2°C. for the duration of the period. The pressure within the chamber shall be maintained at 80 + 2 psi throughout the period. At the completion of the period, the chamber pressure shall be reduced to atmospheric pressure at a uniform rate over a 5-minute period.

4.8.22.2.3 Oxygen pressure. Oxygen pressure aging shall be accomplished by suspending the specimens in an atmosphere of oxygen at 300 ±10 psi for the required period and at the required temperature (see table XIV). The pressure chamber temperature shall be controlled to within plus or minus 1°C. and shall be automatically recorded during the entire period. At the completion of the period, the chamber pressure shall be reduced to atmospheric pressure at a uniform rate over a 5-minute period.

4.8.22.2.4 Hot oil immersion. Specimens shall be immersed in lubricating oil conforming to Symbol No. 2190 TEP of MIL-L-17331, at a temperature and period specified (see table XIV) of 121° ±1°C. for 18 hours. The specimens shall then be removed, blotted lightly to remove excess oil, then suspended in air at room temperature for not less than 3-1/2 nor more than 4-1/2 hours before testing for tensile strength and elongation.

4.8.22.3 Tensile strength and elongation.

4.8.22.3.1 Apparatus. The testing machine shall be power-driven and preferably be the pendulum type. A spring balanced type of apparatus may be used if provided with a device which will indicate the actual maximum load at which rupture takes place and provision is made to prevent recoil of the spring. The testing machine shall be accurate within 1 percent of the breaking load.

4.8.22.3.2 Procedure. The specimen shall be placed in the jaws of the machine and adjusted so that the tension will be distributed uniformly over the cross section. For specimens with bench marks 1 inch apart, the maximum distance between jaws shall be 2 inches. For specimens with bench marks 2 inches apart, the maximum distance between jaws shall be 4 inches. Each specimen shall be stretched at a uniform rate of 20 inches per

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minute until rupture. The elongation shall be measured at the instant of rupture. Rupture of the specimen at any point other than between the bench marks shall invalidate the test. The tensile strength shall be determined from the area of the unstretched and unaged specimen.

4.8.22.3.3 Observation. Failure of the material to meet the tensile and elongation requirements of the specification sheet shall be cause for rejection.

4.8.22.4 Set. The set test shall be performed on unaged specimens which have bench marks 2 inches apart. The specimen shall be placed in the jaws of the machine and adjusted so that the tension will be distributed uniformly over the cross section. The maximum distance between jaws shall be 4 inches. The specimen shall be stretched at a uniform rate of 20 inches per minute until the bench marks are 6 inches apart. The specimen shall be held in the stretched position for 5 seconds and then released. The distance between bench marks shall be determined 1 minute after release and this distance, less 2 inches, shall be considered the set.

4.8.22.4.1 Observation. Failure of the material to meet the set requirements of the specification sheet shall be cause for rejection.

4.8.23 Pressure cycling.

4.8.23.1 Specimen. A specimen shall consist of a 50 to 70-foot length of completed cable or cord.

4.8.23.2 Procedure. The specimen shall be installed in a pressure chamber with both ends brought out approximately 1 foot through bulkhead stuffing tubes conforming to MIL-S-24235/1 and MIL-S-24235/2. The protruding ends of the specimen shall be prepared for capacitance and insulation resistance testing. The specimen within the chamber shall be subjected to five pressure cycles at the pressure specified on the specification sheet. The pressure within the chamber shall be reduced to zero between cycles. For the first and fifth cycles the maximum pressure shall be held for 30 minutes. The maximum pressure for the second, third and fourth cycles shall be held only long enough for the electrical tests to be made. Capacitance and insulation resistance measurements shall be made before and after the pressure cycling and just before pressure is released at the end of each cycle.

4.8.23.3 Observation. The insulation resistance and capacitance values obtained throughout the test shall meet those required by the specification sheet.

4.8.24 Specific gravity. The specific gravity of extruded silicone rubber insulation shall be determined in accordance with method D of ASTM-D297.

4.8.25 Stress endurance.

4.8.25.1 Specimen. The specimen shall consist of a 7-foot length of completed cable.

4.8.25.2 Procedure. The specimen shall be drawn back and forth under alternate loads of 490 and 180 pounds tension through 90 degrees around a 3-inch diameter pulley with a total travel of 18 inches in one direction, for a total of 500 cycles. After the required number of cycles, the tension of the specimen shall be released, and the specimen shall be subjected to the voltage withstand test applicable to the type of cable being tested.

4.8.25.3 Observation. After the stress endurance and voltage withstand tests, the specimen shall be visually examined. Any dielectric failure, breaking of conductors, or cracking of the insulation or jacket shall constitute a failure.

4.8.26 Tension.

4.8.26.1 Specimen. The specimen shall consist of a 6-foot length of completed cable.

4.8.26.2 Procedure. The specimen shall be secured at each end with a woven cable grip similar to type HD of MIL-G-160. A tension load as required by the applicable specification sheet shall be applied to the specimen through the cable grips for 1 hour, and then released. After release of the tension, the specimen shall be given the voltage withstand test applicable to the type of cable being tested.

4.8.26.3 Observation. The specimen shall withstand the tension and voltage withstand tests.

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4.8.27 Thermoplastic flow.

4.8.27.1 Specimen. The specimen shall consist of a length of completed cable sufficient to make two complete turns around the appropriate size mandrel, stripping of ends, and securing to mandrel.

4.8.27.2 Procedure. Ends of specimen shall first be prepared for electrical connections as required herein. The specimen shall then be wrapped for two close turns around a mandrel having a diameter of 10 to 12 times the diameter of the specimen. The specimen shall be secured to the mandrel and placed in an oven. Electrical connections shall be made so that all conductors will be tested from conductor to shield at 70 volts and also between conductors at 120 volts, when two or more conductors have a common shield. The specimen shall be kept in the oven for 6 hours at a temperature of 120° ±2°C. during which time the voltage shall be applied to all conductors.

4.8.27.3 Observation. Electrical failure between conductors or between conductors and shield shall constitute failure of the specimen.

4.8.28 Twisting endurance.

4.8.28.1 Specimen. The specimen shall consist of one length of completed cable or cord. The length shall be 24 inches for cables or cords having a diameter of 5/8-inch or less, and 18 inches plus 10 times the cable diameter in length for cables having a diameter larger than 5/8 inch.

4.8.28.2 Procedures. The specimen shall be conditioned to the temperature specified by the specification sheet and maintained at that temperature during the test. The twisting apparatus may be an assembly powered by a single drive and equipped with means for counting the number of cycles impressed upon the specimen. The specimen shall be clamped in cable grips and subjected to 180 degrees of twist in each direction (360 degrees total twist) at the rate of 12 to 14 complete cycles per minute. The distance between grips shall be 6 inches plus 10 times the maximum diameter of the specimen. The upper grip shall be oscillated by the twisting apparatus. The lower grip shall be free to move vertically, but restrained from oscillation, and shall have a weight attached which stresses the specimen in tension 40 +2 psi of its cross-sectional area. The cross-sectional area of the specimen shall be determined by the maximum diameter for the cable or cord type and size as given by the specification sheet. After completion of the number of cycles and at the temperature specified by the specification sheet the specimen shall be removed from the test apparatus and subjected to the voltage withstand test specified by the specification sheet.

4.8.28.3 Observation. After the twisting and voltage withstand tests, the specimen shall be visually examined. Any voltage withstanding failure, breaking of the conductors, or cracking of the insulation or jacket shall constitute a failure. If failure occurs within 2 inches of the grip and either end of the specimen, the results shall be disregarded and the test repeated.

4.8.29 Visual and dimensional examination. Each selected sample of completed cable or cord shall be examined visually to determine conformance with the requirements of this specification and the applicable specification sheet.

4.8.29.1 Materials examination. Each component material included in the construction of the cable or cord shall be examined to determine that it is an approved material, as shown on the applicable details of Construction sheet (when applicable) issued to the manufacturer by NAVSEC.

4.8.29.2 Dimensional examination. Measurements shall be made to determine conformance with all dimensional requirements of this specification and the specification sheet.

4.8.29.3 Identification. Examination shall be made for proper marker tapes and threads and for proper conductor and group identification codes and methods and surface markings.

4.8.29.4 Repairs. Repairs of any nature to the cable or cord shall be examined for workmanship. Repairs to insulation and jacket shall meet the criteria of 3.7.

4.8.29.5 End seals. End seals shall be examined for proper application and adequacy.

4.8.30 Watertightness.

4.8.30.1 Specimen. A specimen shall consist of a 5-foot length of completed cable.

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4.8.30.2 Procedure. One end of the specimen shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to the water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 psi for a period of 6 hours.

4.8.30.3 Observation. Water leakage through the specimen in excess of that permitted by the specification sheet or table XV, whichever is applicable, shall be cause for rejection.

Table XV - Limits for water leakage.

Total circular-mil area of conductors in cable	Maximum water leakage in cubic inches (cu. in.) during 6-hour test at 25 psi			
	Full contract price (cu. in.)	95 percent of contract price (cu. in.)	90 percent of contract price (cu. in.)	80 percent contract price (cu. in.)
0 to 9,000	1	2	3	4
9,050 to 15,000	2	3	4	5
15,050 to 25,000	3	4	5	6
25,050 to 50,000	4	5	6	8
50,050 to 100,000	5	6	7	9
100,050 to 200,000	6	7	8	10
200,050 to 500,000	7	8	9	11
500,050 to 800,000	8	9	10	12
800,050 to 2,100,000	9	10	11	13

Note 1. - Circular-mil area to be based on nominal circular-mil area required by the specification.
Note 2. - Cable with water leakage in excess of the values shown in the 80 percent column will not be acceptable to the Government.

4.9 Tests methods (electrical).

4.9.1 Attenuation. Attenuation per unit length is defined as the logarithmic decrement in transmitted power. The attenuation, expressed in db per 100 feet, shall be measured at a sufficiently low-power level that the resulting temperature rise will be negligible. An acceptable method for measuring attenuation is as follows:

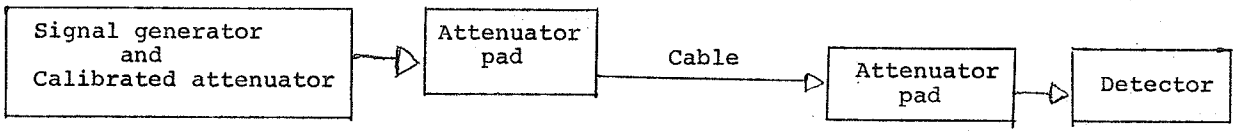


Figure 1 - Attenuation measurement.

In the block diagram (figure 1) a length of cable with an attenuation of at least 3 db shall be inserted between the connectors. The signal generator and calibrated attenuator shall be adjusted to produce a reasonable indication at the detector when the detector is tuned. The detector reading is noted, and the calibrated attenuator output level is recorded. The cable under test is then withdrawn and the circuit completed with the connectors (or a very short length of cable). With the detector tuned, the calibrated attenuator shall be readjusted to reproduce the original reading at the detector, and the attenuator output level is again recorded. Attenuation is then computed as follows:

$$A = \frac{100}{L} \text{ (difference in calibrated attenuator readings in db)}$$

Where:

A = Attenuation in db per 100 feet.
L = Length of cable under test in feet.

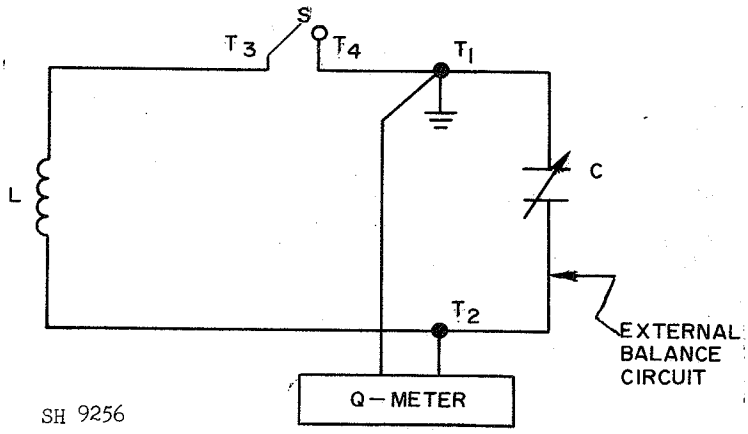
For measurements at frequencies of 400 MHz. or less, the characteristic impedance of the attenuator pads and connectors shall preferably be the same as that of the cable under test.

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4.9.2 Capacitance.

4.9.2.1 Specimen. Specimen shall consist of a 10-foot 2-inch length of completed cable with all shields removed for a distance of 1 inch from each end and the insulation removed from a distance of 1/2 inch from each end of all conductors. The length of the specimen shall be the shielded length.

4.9.2.2 Test equipment. A Q-meter and oscillator equipped with the external balance circuit, as shown on figure 2 is recommended. Other equipment may be used where it can be demonstrated that such an alternate will yield equally accurate results.



L: Coil, of inductance suitable for the magnitude of the reactances to be measured.
C: Variable calibrated precision condenser.
T_1 and T_2: Terminals of Q-meter-T_1 grounded.
T_3 and T_4: Terminals in the inductive branch of the balance circuit. (T_4 can be identical with terminal T_1.)
S: Means of connecting T_3 to T_4.

Figure 2 - Test equipment for capacitance test.

The leads from L and C to the Q-meter terminals shall be as short as practicable. The test specimen or leads from the specimen shall be terminated at or as near as possible to the terminals of L.

4.9.2.3 Procedure (shielded singles). The shield at each end of the conductor under test shall be grounded. Measurements shall be made for each shielded conductor using the frequency required by the specification sheet. Proceed as follows for each conductor of the specimen:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Connect the conductor to terminal T_2 and the shield to terminal T_1. Balance the circuit by adjusting C to a new value. Designate this value as C_1.

4.9.2.3.1 Calculation. The capacitance (C) per foot, for each conductor, shall be determined by the formula:

$$C = \frac{(C_0 - C_1)}{(\text{length of specimen in feet})}$$

4.9.2.4 Procedure (shielded pairs). All shields at each end of the specimen shall be grounded. The capacitance shall be measured for each pair at the frequency specified on

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the specification sheet. For each pair, one conductor shall be designated as No. 1, the other conductor as No. 2 and the shield as No. 3. Proceed as follows for each pair:

(a) Connect terminal T_3 to T_4.

(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_o.

(c) Connect No. 1 and No. 3 to terminal T_1, and connect No. 2 to terminal T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance $(C_o - C_1)$ measured between terminals T_1 and T_2 as C_a.

(d) Disconnect No. 1 from terminal T_1 and No. 2 from terminal T_2. Connect No. 2 with No. 3 to terminal T_1 and connect No. 1 to terminal T_2. Balance the circuit by adjusting C to a new value C_2. Designate the capacitance $(C_o - C_2)$ as C_b.

(e) Disconnect No. 2 from terminal T_1 and connect it to terminal T_2. Balance the circuit by adjusting C to a new value C_3. Designate the capacitance $(C_o - C_3)$ as C_c.

4.9.2.4.1 Calculation. Calculation shall be made as follows:

(a) The mutual capacitance (C_m) per foot for each pair shall be determined by the formula:

$$C_m = \frac{2(C_a + C_b) - C_c}{4X(\text{Length of specimen in feet})}$$

(b) The capacitance unbalance or coefficient of asymmetry (C_u) of a shielded pair, expressed in percent, shall be determined by the formula:

$$C_u = \frac{400(C_a - C_b)}{2(C_a + C_b) - C_c}$$

4.9.2.5 Procedure (shielded triads). The shield at each end of the triad under test shall be grounded. Measurements shall be made for each triad using the frequency required by the specification sheet. For each triad designate one conductor as No. 1, the second conductor as No. 2; the third conductor as No. 3 and the shield as No. 4. Proceed as follows for each triad:

(a) Connect terminal T_3 to T_4.

(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_o.

(c) Connect No. 2, No. 3 and No. 4 to terminal T_1. Connect No. 1 to terminal T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance $(C_o - C_1)$ measured between terminals T_1 and T_2 as C_a.

(d) Disconnect No. 2 from terminal T_1 and disconnect No. 1 from terminal T_2. Connect No. 2 to terminal T_2. Connect No. 1 with No. 3 and No. 4 to terminal T_1. Balance the circuit by adjusting C to a new value C_2. Designate this capacitance $(C_o - C_2)$ as C_b.

(e) Disconnect No. 3 from terminal T_1 and disconnect No. 2 from terminal T_2. Connect No. 3 to terminal T_2. Connect No. 1 with No. 2 and No. 4 to terminal T_1. Balance the circuit by adjusting C to a new value C_3. Designate this capacitance $(C_o - C_3)$ as C_c.

(f) Disconnect No. 1 and No. 2 from terminal T_1 and connect them with No. 3 to terminal T_2. Balance the circuit by adjusting C to a new value C_4. Designate this capacitance $(C_o - C_4)$ as C_d.

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4.9.2.5.1 Calculation. Calculation shall be made as follows:

(a) The mutual capacitance (C_m) per foot for each shielded triad shall be determined by the formula:

$$C_m = \frac{3\ (C_a + C_b + C_c)\ - C_d}{12 \times (\text{length of specimen in feet})}$$

(b) The capacitance unbalance or coefficient of asymmetry (K) of a shielded triad, expressed in percent, shall be determined by the formula:

$$K_1 = \frac{200\ (C_a - C_b)}{(C_a + C_b)}$$

$$K_2 = \frac{200\ (C_a - C_c)}{(C_a + C_c)}$$

$$K_3 = \frac{200\ (C_b - C_c)}{(C_b + C_c)}$$

Where: K_1 = The percent of capacitance unbalance of conductor No. 1 in relation to conductor No. 2.
K_2 = The percent of capacitance unbalance of conductor No. 2 in relation to conductor No. 3.
K_3 = The percent of capacitance unbalance of conductor No. 3 in relation to conductor No. 1.

Note: In the formula, the relative positions of C_a, C_b and C_c may be transposed as necessary to avoid negative values.

4.9.2.6 Procedure (unshielded pairs). For unshielded pairs, the procedure shall be the same as given for shielded pair cables except the overall shield, if any, and all conductors of the specimen, except the pair under test shall be designated and treated as No. 3 when making measurements.

4.9.2.6.1 Calculations. The mutual capacitance (C_m) for each pair shall be determined by the formula as given for shielded pairs.

4.9.2.7 Procedure (unshielded triads). For unshielded triads the procedure shall be the same as for triads, except that the overall shield if any, and all conductors of the specimen outside of the triad under test, shall be designated and treated as No. 4 when making measurements.

4.9.2.7.1 Calculations. The mutual capacitance (C_m) and the capacitance unbalance (K) shall be determined by the formulas specified for shielded triads.

4.9.3 Characteristic impedance. The characteristic impedance for shielded singles or paired conductors shall be determined from the measured values of capacitance and the inductance.

4.9.3.1 Specimen. The specimen shall consist of a 10-foot 2-inch length of completed cable with all shields removed for a distance of 1 inch from each end and the insulation removed for a distance of 1/2 inch from each end of all conductors. The length of the specimen shall be the shielded length.

4.9.3.2 Procedure. Measurements for the capacitance shall be made first in accordance with the procedure given for the cable construction being tested. Measurements for determining the inductance shall be made as follows and on the same specimen used for determining the capacitance. (See 4.9.2).

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4.9.3.2.1 Measurements for shielded pairs shall be made as follows:

(a) Connect terminal T_3 to T_4.

(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_O.

(c) Disconnect terminal T_3 from T_4. Connect the two conductors of the pair under test together at the far end of the specimen. Connect conductor No. 1 to terminal T_3. Connect conductor No. 2 and the shield of the pair to terminal T_4. Balance the circuit by adjusting C to a new value C_1. Designate the change $(C_O - C_1)$ as dC.

4.9.3.2.2 Measurements for shielded singles shall be made as follows:

(a) Connect terminal T_3 to T_4.

(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_O.

(c) Disconnect terminal T_3 from T_4. Connect the shield to the conductor under test at the far end of the specimen. Connect the conductor to terminal T_3 and connect the shield to terminal T_4. Balance the circuit by adjusting C to a new value C_1. Designate the change $(C_O - C_1)$ as dC.

4.9.3.2.3 Calculations. The inductance of the shielded pair or shielded single (L_p) may be determined by the formula:

$$L_p = \frac{dC}{W^2 C_O (C_O - dC)}$$

Where W = 2π times the frequency. If dC is 1 percent or less of C_O, the formula may be reduced to:

$$L_p = \frac{dC}{W^2 C_O^2}$$

The characteristic impedance of the shielded pair or shielded single (Z_O) may be determined by the formula:

$$Z_O = \sqrt{\frac{L_p}{Cp}}$$

4.9.4 Conductor resistance. Electrical resistance shall be determined on conductors of completed cable. The test procedure shall be in accordance with ASTM B193. The measured resistance shall be not greater than the value permitted by the specification sheet.

4.9.5 Insulation resistance. The insulation resistance shall be determined for conductor insulation and for cable or cord jackets, when required by the specification sheet.

4.9.5.1 Procedure. The test shall be performed on each length of completed cable or cord immediately following the voltage withstand test. The leakage current shall be measured after 1-minute electrification with a direct current potential of not less than 200 nor more than 500 volts. Multi-conductor cables or cords with unshielded conductors shall be tested between each conductor and all other conductors to overall shield, armor or water. Cables or cords having shielded groups, such as pairs and triads, shall be tested between each conductor and all other conductors in the group and to the group shield. Cables or cords with individually shielded conductors shall be tested between conductor and shield. For cables or cords having combinations of shielded and unshielded, groups or singles, the applicable procedure shall be used. Where cable or cord jackets have insulation resistance requirements, the test shall be made between the overall cable or cord shield and the water bath. The conductor or shield whose insulation is under test shall be connected to the negative terminal of the test equipment and readings shall be taken after 1-minute electrification.

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4.9.5.2 Observation. The insulation resistance values at 15.5°C. shall be not less than required by the specification sheet. If the measurement is made at a temperature other than 15.5°C., the manufacturer shall correct the measured value to 15.5°C. If the insulation resistance is equal to or greater than that required, when the measurement is made at a temperature greater than 15.5°C., no correction factor need be employed. The manufacturer shall demonstrate that the correction factor used is accurate for his insulating compound.

4.9.6 Mutual inductance.

4.9.6.1 Specimen. The specimen shall consist of a 10-foot length of completed cable with the insulation removed for a distance of 1/2 inch from each end of all conductors. The braided copper shield shall be removed or pushed back to expose 1-1/2 inches of the twisted pair at both ends. The conductors of one end of each shielded pair shall be shorted.

4.9.6.2 Procedure. An approximate 1000-Hz current shall be circulated in the red and green conductors, then in the red and blue conductors, then in the green and blue conductors for MCOS-5, and in the yellow and green conductor for MCOS-6. The induced voltage appearing across the open end of each shielded pair shall be measured to an accuracy of 1 microvolt while the current is flowing.

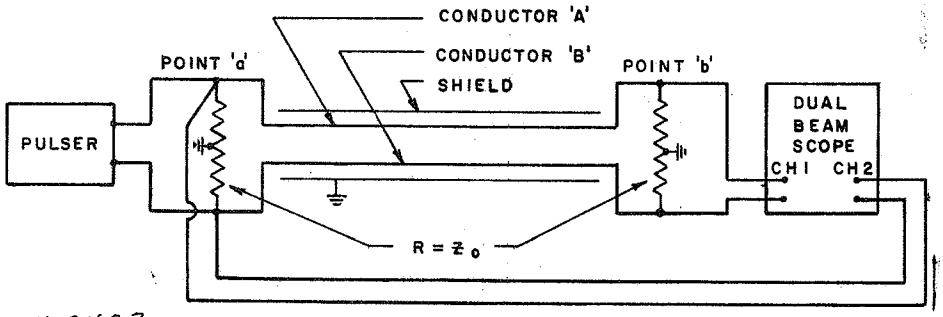
4.9.6.3 Calculations. The mutual inductance in henries shall be determined as follows:

$$\text{Mutual inductance} = \frac{\text{Induced voltage (microvolts)}}{2\pi \times \text{frequency (Hz)} \times \text{current (amps)}}$$
Mutual

4.9.7 Pulse response time. The following procedure shall be used for the determination of the pulse response time:

(a) The test shall be performed on the reel length of a balanced cable circuit. The balanced condition can be achieved by several means:
(1) Employing balanced test equipment.
(2) Transformer - These must be good quality pulse transformers.
(3) Electronic balancing circuits or devices.

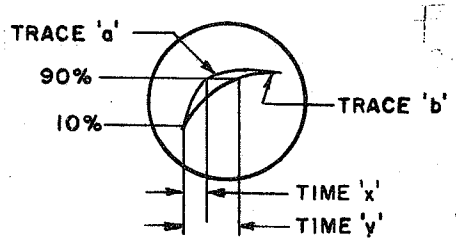
A dual beam oscilloscope of the fast rise type with a good band pass capability combined with a good quality square wave pulser connected as shown in schematic in figure 3 (simplified representative schematic).



SH 8423

Figure 3 - Pulse response circuit diagram.

(b) It is important that the generator output pulse width be of sufficient duration to permit trace "b" of figure 4 to reach its full amplitude before "cutoff" is reached. For this test a pulse duration of 2.0 micro-seconds will be used.
(1) Rise time determination. The dual beam oscilloscope should show a trace with the following characteristics:



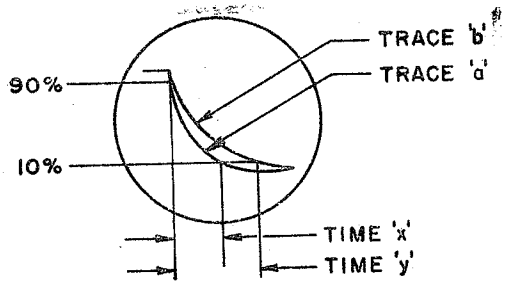
SH 8424

Figure 4 - Rise time.

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Time "x" is the measurement of the arithmetic mean of the times of all parts (pulser, scope, test leads to point "a", and so forth) connected to the input end of the cable. Time interval "y" is the mean of time "x" combined with the cable's rise time. It is important that the length, composition, and characteristics of the connecting leads to the oscilloscope from points "a" and "b" be identical, and as short as possible. Now, with both traces on the same time base, and coincident at the 10 percent amplitude value, determine the time duration of trace "b" from the 10 percent to 90 percent amplitude. Call this time "y". Next, insert these values ("x" and "y") into the formula $t_r = \sqrt{y^2 - x^2}$ where t_r is the cable's rise time and "x" and "y" are the times previously determined.

(2) Decay time (see figure 5) may be determined as follows:



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Figure 5 - Decay time.

Traces "a" and "b" are coincident at the 90 percent amplitude value. With the traces on the same time base, time interval "x" is the decay time mean as in the rise time determination, and time interval "y" is the mean of all decay times including that of the cable. As before, insert these values of "x" and "y" into the formula:

$$t_d = \sqrt{y^2 - x^2}$$

Where: t_d = the decay time of the cable: "x" and "y" are as previously described.

(c) If transformers or other balancing devices are used to provide the balanced condition, time "x" will have to be determined by reading the time of trace "b" in both cases with the cable out of the circuit, and the transformers or devices connected directly together, then time "y" will be determined as before with the cable inserted in the circuit. The difference between the traces "a" and "b" when the transformers or devices are connected directly together is the difference in the mean value of the rise and decay time caused by these items. One channel of the dual beam scope is connected at point (a) in this test to provided a reference for any possible change to the input pulse caused by the insertion of the cable.

(d) Insert the values found for the rise and decay times into the formulas:

$$t_r/500 = \frac{500\ t_r}{l}$$

$$t_d/500 = \frac{500\ t_d}{l}$$

to determine the rise or decay time for a 500 ft. length of cable.

Where: t_r = rise time determined for sample.

t_d = decay time determined for sample,

l = sample length in feet.

4.9.8 Voltage withstand. Voltage withstand tests shall be made on all lengths of completed cable or cord.

4.9.8.1 Apparatus. The voltage withstand tests shall be made with alternating potential from a source of ample capacity, but in no case less than 5 kilovolt-ampere, having

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a frequency not greater than 100 Hz and a wave shape approximately a sine wave under all test conditions. The testing voltage may be measured by any method satisfactory to NAVSEC, which gives a root-mean-square value. The preferable method is by means of a voltmeter connected to voltmeter coil in the high-tension winding of the testing transformer, or to a separate instrument transformer.

4.9.8.2 Procedure. The test voltages and application (conductor to ground, conductor to conductor, shield to shield, and so forth) shall be as required by the specification sheet. The time of application for all voltage withstand tests shall be 1 minute. The initially applied voltage shall be not greater than 600 volts. The rate of increase shall be approximately uniform and not over 100 percent in 10 seconds nor less than 100 percent in 60 seconds. All unarmored cables requiring electrical tests for the jacket shall be immersed in a grounded water bath for at least 1 hour, and tested while still immersed, using the water as the ground.

4.9.8.3 Observation. All cable shall withstand without failure the voltages specified on the specification sheet.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 All cables covered by this specification shall be shipped on reels, unless otherwise specified on the specification sheet.

5.1.1 Reels. Reels shall be of a design, strongly constructed, with flanges of sufficient diameter to protect the cable from injury during handling and storage. Where the gross weight of cable and reel exceed 1000 pounds, the reels shall have metal hubs. The diameter of the drum shall be not less than 14 times the overall diameter of the cable.

5.1.1.1 Placement of cable on the reel. The cable on each reel shall have both ends readily available for testing without re-reeling. The inner end shall extend through a hole in the flange near the drum, and the outer end shall be available at the top of the reel. Both ends of the cable shall be secured to the reel flange with 1/4-inch rope, applied with two half-hitches, one 6 inches and the other 4 inches (approximately) back of the end seal, then twisting the rope ends together, knotting them, pulling the cable up tight and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange. The inner end which extends through the flange hole shall be protected by a metal end cover.

5.1.1.2 Lagging. Lagging shall be used on reels having flange diameters of 42 inches and larger and on all reels which contain 1000 pounds or more of cable. Reels smaller than 42 inches in diameter which contain less than 1000 pounds of cable may be covered in accordance with 5.1.1.3. Lagging boards shall have a thickness equal to the thickness of the reel flange but not greater than 2-inch commercial lumber. Lagging for all reels shall extend to within 1/4-inch of the outside of the flanges. All lagging boards shall be installed so that they are touching adjacent boards. All lagging which rests entirely on the rim of the flange shall be nailed with cement-coated nails. Lagging which is notched and recessed may be secured by the use of strapping only. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing 1000 pounds or more and having a flange diameter less than 78 inches , two 0.162 ±0.003-inch (8-gage) high-tensile round steel wires or two 3/4 by 0.035-inch steel straps shall be used. For all reels having a flange diameter of 78 inches or more two 1-1/4 by 0.035-inch steel straps shall be used. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing less than 1000 pounds, two 0.0915 ±0.003-inch (13-gage) high-tensile round steel wires or two 5/8 by 0.020 inch steel straps shall be used. All straps shall be stapled at approximately 15-inch intervals. A minimum clearance of 2 inches shall be provided between inner face of lagging and outer layer of cable.

5.1.1.3 Alternate to lagging. As an alternate to lagging for reels having a flange diameter smaller than 42 inches and containing less than 1000 pounds of cable, a covering may be used which consists of the following:

(a) A cushioning inner layer of asphalt-impregnated, corrugated, fiberboard, or the equivalent, wrapped to give 100-percent coverage.
(b) An outer covering consisting of two layers of weather-resistant veneer paper conforming to PPP-V-205 or two layers of three-ply weather-resistant toxic-

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treated plywood wrapped to give 100-percent coverage.

This covering shall be secured in place between the reel flanges by at least two 3/8 by 0.020-inch steel straps.

5.1.1.4 Yearly marking. In addition to any other required marking, reels shall be marked with a keyed series of colors to indicate the year of manufacture. The cycle of colors shall be repeated every fifth year. The reel marking shall consist of a stripe approximately 2 inches wide, colored to designate the particular year of manufacture as follows:

Year of manufacture		Identifying color
1971	1976	Red
1972	1977	Green
1973	1978	Orange
1974	1979	Blue
1975	1980	White

5.1.1.4.1 Location of yearly marking. The colored stripe on reels shall be applied circumferentially over the lagging or the alternate to lagging and midway between the flanges. The stripe shall consist of one coat of commercial quality outside paint of the appropriate color. In addition to the stripe, both flanges of the reel shall be stenciled with 4-inch high figures to show the year of manufacture.

5.1.1.5 Standard reel markings. Each reel shall be plainly marked on both flanges with the following information:

(a) Reel number.
(b) Type and size of cable.
(c) Footage.
(d) Contract or order number.
(e) Contractor's name.
(f) Manufacturer's name (if other than contractor).
(g) Gross weight.

5.1.2 Coils. Coils shall contain one continuous length of cable or cord (standard length, see table XVI).

5.1.2.1 Yearly marking. In addition to any other required marking, coils shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2-inches wide and colored for the particular year of manufacture. The cycle of colors shall be the same as those used for yearly marking of reels (see 5.1.1.4).

5.1.2.2 Identification. Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information:

(a) Type and size of cable or cord.
(b) Footage.
(c) Contract or order number
(d) Contractor's name.
(e) Manufacturer's name (if other than contractor).
(f) Gross weight.

5.2 Packaging and packing. Cable and cord as specified (see 6.2) shall be packaged and packed level A, B, or C in accordance with MIL-C-12000, as specified (see 6.2). Marking shall be in accordance with MIL-C-12000 and with the requirements of any special marking specified herein or as specified (see 6.2).

6. NOTES

6.1 Intended use. Cables and cords specified herein are intended for use in various application involving naval ships and shore stations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of cable or cord required with reference to applicable specification sheet.

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(c) Quantity required.
(d) Level of packaging and packing required (see 5.2).
(e) Special marking required (see 5.2).

6.2.1 Quantity. The quantity of each type and size on a contract or order should be specified as an integral multiple of the unit ordering length shown by the specification sheet.

6.2.2 Lengths. The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XVI. In order to compensate for handling and probable loss on issue, remnant lengths are subject to the price reductions given in table XVI. Scrap lengths are not acceptable. It is not intended that items being procured in exact lengths for a specific job or ship shall necessarily be shipped in lengths which are integral multiples of the unit ordering length or that price reductions shall apply in the event that exact footages or fractional lengths are required and so indicated in procurement of cable or cord for other than stock purposes.

Table XVI - Cable or cord lengths.

Type of package	Nominal length (feet)	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
		No price reduction (feet)	No price reduction (feet)	Price reduction		Not acceptable (feet)
				5 percent (feet)	10 percent (feet)	
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1500	1650 to 1350	1349 to 450	449 to 300	299 to 150	149 to 0
Reel	2000	2200 to 1800	1799 to 600	599 to 400	399 to 200	199 to 0
Reel	2500	2750 to 2250	2249 to 750	749 to 500	499 to 250	249 to 0
Reel	3000	3300 to 2700	2699 to 900	899 to 600	599 to 300	299 to 0

6.3 Special clause. Except when small quantities are to be purchased, invitations for bid and contracts or orders should specify the following:

"In order to compensate for handling and probable loss on issue, remnant lengths shall be subject to the price reductions shown in table XVI. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable or cord on the contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximately two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in any combination of random lengths."

6.4 Time delay. Comparison inspection will normally require 40 calendar days from the date the sample is sent to the Government-designated laboratory to the date the inspection results are received by the manufacturer. This time delay should be taken into consideration by a supplier when estimating delivery time.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-915 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with 6.5.1 and "Provisions Governing Qualification SD-6" (see 6.5.2).

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6.5.1 Application for qualification. The application shall provide the following information:

(a) Company name and business address. Also address of manufacturing facility if different from business address.
(b) The type and size designations of the cables or cords for which qualification approval is being sought.
(c) A details of construction sheet covering each type of cable or cord for which qualification approval is being sought. This details of construction shall show all materials by manufacturer's name and part number which the cable or cord manufacturer proposes to use in the construction of his product.
(d) If the company proposes to have the qualification testing performed at a laboratory other than his own, the name and address of the laboratory.

6.5.2 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material is procured by the supplier for incorporation into the cable or cord and loses separate identity when the cable or cord is shipped.

6.7 Supersession data. This specification supersedes Specifications MIL-C-00915D(SHIPS), MIL-C-915C of 15 March 1968, MIL-C-2194E(SHIPS) of 31 January 1968, MIL-C-23206(SHIPS) of 23 February 1962 and MIL-C-24145A(SHIPS) of 16 October 1967.

6.7.1 New types and obsoleted types. All detail specifications covered by the specifications in 6.7 are listed in table XVII both by type designation and by detail specification number. Table XVII also lists (a) new type designations for cables for which a design change has been made, and a remark on the nature of the design change; (b) the replacement types, for cable types which have been obsoleted.

Table XVII - Supersession data

Previous type and detail specification number	Present type and applicable MIL-C-915 Specification sheet number	Remarks
CVSF MIL-C-915/1	CVSF ---- /1	No change
DBSP /2	2SJ ----- /60	DBSP obsolete
DCOP /3	DCOP----- /3	No change
DHFR /4	DRW ----- /73	DHFR obsolete
DHOF /6	DHOF----- /6	No change
DLT MIL-C-915/5	DLT ----- /5	No change
DPS MIL-C-23206/3	DPS ----- /40	No change
DSGA MIL-C-2194/2	DSGU ---- /29	Armor eliminated
DSS MIL-C-915/8	DSS ----- /8	No change
DSWS /7	DSWS ---- /7	No change
ECM MIL-C-24145/14	ECM ----- /54	No change
FBSP MIL-C-915/2	4SJ ----- /60	FBSP obsolete
FHOF /6	FHOF ---- /6	No change
FPS MIL-C-23206/3	FPS ----- /40	No change
FSGA MIL-C-2194/4	FSGU ---- /31	Armor eliminated
FSS MIL-C-915/8	FSS ----- /8	No change
JAS /9	JAS ----- /9	No change
MA MIL-C-24145/3	MU ------ /43	Armor eliminated
MCOS MIL-C-915/11	MCOS ---- /11	No change
MCSF /10	MCSF ---- /10	No change

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Table XVII - Supersession data (cont'd).

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
MDU MIL-C-915/12	MDU ----- /12	No change
MDY /13	MDU ----- /12	MDY obsolete
MHOF /14	MHOF ---- /14	No change
MMOP /15	MMOP ---- /15	No change
MRI /16	MRI ----- /16	No change
MS MIL-C-24145/10	MS ------ /50	No change
MSCA MIL-C-2194/5	MSCU ---- /32	Armor eliminated
MWF MIL-C-24145/18	MWF ----- /58	No change
PBTM MIL-C-915/17	PBTMU --- /17	Armor eliminated
PI MIL-C-23206/2	PI ------/39	Improved design
SHFR MIL-C-915/4	SRW ----- /73	SHFR obsolete
SHOF /6	SHOF ---- /6	No change
SSF /18	SSF ----- /18	No change
SSGA MIL-C-2194/1	SSGU ---- /28	Armor eliminated
SS5P MIL-C-2194/7	None	SS5P obsolete
S2S MIL-C-24145/21	S2S ----- /61	No change
TBSP MIL-C-915/2	3SJ ----- /60	TBSP obsolete
TCJA MIL-C-2194/10	TCJU ---- /35	Armor eliminated
TCJX MIL-C-23206/1	TCJX ---- /38	No change
TCKX /1	TCKX ---- /38	No change
TCOP MIL-C-915/3	TCOP ---- /3	No change
TCTA MIL-C-2194/10	TCTU ---- /35	Armor eliminated
TCTX MIL-C-23206/1	TCTX ---- /38	No change
THFR MIL-C-915/4	TRW ----- /73	THFR obsolete
THOF /6	THOF ---- /6	No change
TPS MIL-C-23206/3	TPS ----- /40	No change
TPU MIL-C-915/19	TPU ----- /19	No change
TRF /20	TRF ----- /20	No change
TRXF /21	TRXF ---- /21	No change
TSGA MIL-C-2194/3	TSGU ---- /30	Armor eliminated
TSP MIL-C-915/22	TSP ----- /22	No change
TSS MIL-C-915/8	TSS -----/8	No change
TTHFWA /23	TTSU --- /37	TTHFWA obsolete
TTOP /24	TTOP ----/24	No change
TTRS /25	TTRS ----/25	No change
TTRSA /26	TTRS ----/25	TTRSA obsolete
TTSA MIL-C-2194/12	TTSU ----/37	Armor eliminated
1SA MIL-C-24145/17	1SAU ----/57	Armor eliminated
1SMWA /27	1SMWU ---/65	Armor eliminated
1S50MA /2	1S50MU --/42	Armor eliminated
1S75MA /15	1S75MU --/55	Armor eliminated
1SMU MIL-C-24145/16	1SMU ----/56	No change
1SU /19	1SU -----/59	No change
1SWA /4	1SWU ----/44	Armor eliminated
1SWF /7	1SWF ----/47	No change

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Table XVII - Supersession data (Cont'd).

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
1SMU MIL-C-24145/16	1SMU ----/56	No change
1SU /19	1SU -----/59	No change
1SWA /4	1SWU ----/44	Armor eliminated
1SWF /7	1SWF ----/47	No change
2A /1	2AU -----/41	Armor eliminated
2SA MIL-C-24145/5	2SU -----/45	Armor eliminated
2SJ /20	2SJ -----/60	No change
2SWA /6	2SWAU ---/46	Armor eliminated
2SWF /8	2SWF ----/48	No change
2SWU /9	2SWU ----/49	No change
2U MIL-C-24145/23	2U ------/63	No change
2WA /24	2WAU ----/64	Armor eliminated
3SA /11	3SU -----/51	Armor eliminated
3SF /22	3SF -----/62	No change
3SJ /20	3SJ -----/60	No change
3SWA MIL-C-24145/12	3SWU ----/52	Armor eliminated
3U /13	3U ------/53	No change
4SJ /20	4SJ -----/60	No change
5KVTSGA MIL-C-2194/11	5KVTSGU--/36	Armor eliminated
6SGA MIL-C-2194/6	6SGU ----/33	Armor eliminated
7PS MIL-C-23206/3	7PS -----/40	No change
7SGA MIL-C-2194/9	7SGU ----/34	Armor eliminated
7SS MIL-C-915/8	7SS -----/8	No change
	DNW -----/68	New type
	FNW -----/70	New type
	MNW -----/71	New type
	MSP -----/67	New type
	MSPW ----/66	New type
	TNW -----/69	New type
	TPNW ----/72	New type
	5SB -----/74	New type

6.8 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Army - MI
Navy - SH
Air Force - 80
Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609)

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